



SOFTWARE ARCHITECTURE PROJECT

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Software engineering Level two

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Phase 1: Analysis

1.1 System Actors and Access Control (RBAC)

1- Client Portal

Primary System Actor:

- Wholesale Client

Granted Access & Privileges:

- Register corporate profile accounts into a pending verification status.
- Manage and update active corporate profile account details.
- Delete corporate profile accounts from the portal.
- Upload multi-item inventory demand sheets in standard spreadsheet formats.
- Use integrated AI tools to read and compare different pricing options.
- View generated quotation sheets compiled from low-rate market options.
- Accept or reject systemic pricing valuations.

Other System Actor:

- System Administrator

Granted Access & Privileges:

- Review pending wholesale client registration profiles to grant portal access.
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2- Admin Dashboard

Primary System Actor:

- Admin

Granted Access & Privileges:

- Create new internal employee or manager accounts.
- Delete internal employee or manager accounts.
- Assign roles and permissions to internal staff members to enforce access control (RBAC).
- Review and approve pending wholesale client profiles to activate portal features.

Other System Actor:

- Manager and Employee

Granted Access & Privileges:

- Monitor active order lifecycles throughout the procurement process.
 - Track general platform processes and system workflows.
 - Publish manual administrative system adjustments directly to the central event bus.
 - Flag suspicious client profiles for administrative review.
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3- Integration Gateway

Primary System Actor:

- Odoo ERP System

Granted Access & Privileges:

- Synchronize accounting ledgers, wholesale billing details, and central enterprise records.

Other System Actor:

- Google Maps API

Granted Access & Privileges:

- Provide geospatial verification coordinates for domestic and international warehouses.

Other System Actor:

- WhatsApp Gateway API

Granted Access & Privileges:

- Transmit automated text notifications and purchase confirmations to corporate clients.

Other System Actor:

- Third-Party Logistics APIs

Granted Access & Privileges:

- Feed live transit milestones and updates directly into the shipment monitoring framework
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4- Local Supplier Portal

Primary System Actor:

- Local Supplier

Granted Access & Privileges:

- Create a vendor profile account on the platform.
- Manage corporate vendor account details.
- Delete the vendor profile account.
- Add new product items to their corporate stock list.
- View personal warehouse product items and current stock lines.
- Edit existing product prices and localized pricing tiers.
- Delete items from their online stock list.
- Receive instant digital purchase orders when clients accept matching quotes.
- Publish stock level updates to the underlying message network to ensure immediate synchronization

1.2 Functional requirements

1- Client Portal

Client (Main Actor)

- The system allows the user to create a new corporate account profile.
- The system allows the user to log into the portal.
- The system allows the user to manage account profile information.
- The system allows the user to delete their account profile.
- The system allows the user to upload an Excel sheet to request automatic part prices.
- The system allows the user to use integrated AI tools to read and compare different pricing options.
- The system allows the user to view and download a generated PDF quotation sheet.
- The system allows the user to accept or reject the offer.

System Administrator (Secondary Actor)

- The system allows the user to review new registration requests.

- The system allows the user to approve a new client account to give them system access.

2- Admin Dashboard

Admin (Main Actor)

- The system allows the user to create new internal employee or manager accounts.
- The system allows the user to delete internal employee or manager accounts.
- The system allows the user to assign roles and permissions to internal staff members.
- The system allows the user to review and approve pending wholesale client profiles.

Manager (Main Actor) and Employee (Main Actor)

- The system allows the user to monitor active order lifecycles.
- The system allows the user to track general platform processes and system workflows.
- The system allows the user to publish manual system updates to the event bus.
- The system allows the user to flag client profiles for administrative review.

3- Integration Gateway

External Systems (Odoo ERP, Google Maps, WhatsApp, Logistics APIs)

- The system allows the external system to verify secure connection keys before trading any data.
- The system allows the external system to validate all incoming and outgoing data traffic.
- The system allows the external system to translate data formats before they reach the main application core.
- The system allows the external system to convert logistics tracking updates and accounting data into internal system messages.
- The system allows the external system to register as a publisher or subscriber to events on the platform message network.
- The system allows the external system to pass business data back and forth without connecting directly to internal tools.

4- Local Supplier Portal

Local Supplier (Main Actor)

- The system allows the user to create a vendor profile account.
- The system allows the user to manage their corporate account details.
- The system allows the user to delete their vendor profile account.
- The system allows the user to add new product items to their stock list.
- The system allows the user to view their personal warehouse product items.
- The system allows the user to edit existing product prices and current quantities.
- The system allows the user to delete items from their online stock list.
- The system requires all typed supplier inputs to pass automated validation check gates on the server side.
- The system allows the user to receive automated alerts when a buyer accepts a quote.
- The system allows the user to publish a "StockUpdated" event to the message network when stock numbers change.

1.3 External System Requirements & Integrations

1- Odoo ERP System

- **What it does:** It manages the core corporate accounting, business data persistence, and financial transaction records for the platform.
- **Why it is needed:** The platform needs a safe and legal way to handle high-volume wholesale financial transactions. Instead of building a custom accounting tool, the system connects to Odoo to sync invoices, track tax records, and manage financial ledgers automatically when quotes are accepted.

2- Google Maps API

- **What it does:** It handles all geographic mapping, location services, and warehouse coordinate checks.
- **Why it is needed:** Automated part matching requires knowing the exact physical locations of local and international supplier warehouses. The system needs Google Maps to calculate shipping distance, display vendor locations, and help calculate correct logistics costs for the buyer.

3- WhatsApp Gateway API

- **What it does:** It automates instant text notifications, order confirmation updates, and system alerts to user mobile phones.
- **Why it is needed:** Wholesale buyers and suppliers need instant updates to finish deals quickly. Since users do not look at the web portal constantly, this integration sends immediate alerts to their WhatsApp when a new price quote is ready or when an order state changes.

4- Third-Party Logistics APIs

- **What it does:** It connects the platform with external shipping providers to manage delivery tracking and transit data.
- **Why it is needed:** After a wholesale client accepts a quote, physical automotive parts must move from warehouses to workshops. This integration allows the system to ingest real-time shipping milestones so both buyers and managers can see tracking information directly on their dashboards.

5- B2B Payment Gateway API (بوابة الدفع الإلكتروني المخصصة للشركات):

- **What is it?** A service like **Stripe**, **Paytabs**, or **Mada/Sadad** integrated via code.
- **Why we need it:** When a wholesale buyer accepts a PDF quote, they need to pay for the order immediately so the system can notify the suppliers to ship the parts. We cannot have a fully automated system if someone has to manually do bank transfers or pay cash outside the platform.

1.4 Selected Hosting Infrastructure: Amazon Web Services (AWS)

I have decided to fully build and deploy the system on **Amazon Web Services (AWS)**. This platform gives us the exact physical and software tools we need to support our multi-portal marketplace layout safely and efficiently.

Core Architectural Benefits for PartZone B2B

- **Data Residency Inside Saudi Arabia:** This is arguably our biggest advantage. Because AWS has dedicated, official data centers physically located within the Kingdom of Saudi Arabia, we can safely store all wholesale transaction logs, client company files, and financial records inside the country.
- **Incredible Low-Latency Speed:** Since our servers sit right inside the local Saudi region, the physical distance between our servers and our local wholesale buyers or domestic suppliers is minimal. This ensures that uploading huge part request Excel files, generating heavy PDF offers, and processing page loads happens instantly without frustrating network lags.
- **Native Tools for Our Event-Driven Bus:** Our system relies heavily on a Pub/Sub layout where different portals communicate using background . AWS

provides excellent, easy-to-use built-in messaging tools like **AWS SNS (Simple Notification Service)** and **AWS SQS (Simple Queue Service)** out of the box. These tools handle asynchronous background messaging perfectly so our portals never freeze up while waiting for data.

- **Perfect Separation for Decoupled Micro-layers:** Because we are using an N-Tier multi-layered design, we do not want a heavy load on the Supplier Portal to crash the main Client Portal. AWS lets us host the Client Portal, Supplier Portal, and Admin Dashboard on completely isolated server components. They remain completely independent but can talk to each other cleanly over secure data channels.
- **Enterprise Security and Protection:** Dealing with corporate business procurement means we are an attractive target for digital attacks. AWS provides excellent automated defenses like firewalls and traffic filters.

1.5 Suggest the tools and frameworks that we use in the system, alongside Database Suggestion & Vendor Comparison

1.5.1 – The Frameworks

1.5.1.1 - Frontend (What the users see)

- **React.js**
 - **What it is:** A very popular, simple framework used to build modern, interactive user interfaces.
 - **Benefit:** It is perfect for splitting your system into separate portals. It keeps the **Client Portal**, **Supplier Portal**, and **Admin Dashboard** completely independent, fast, and easy to update without breaking each other.

1.5.1.2 Backend (The hidden engine)

- **Node.js (Express)**
 - **What it is:** A fast, lightweight backend framework used to handle web server traffic.
 - **Benefit:** It works seamlessly with automated event networks. It allows the system to process heavy Excel files from clients and immediately send out price update notifications without locking up the user's screen.

1.5.2 - Database Suggestion & Vendor Comparison

The selected database provider for the PartZone platform is **PostgreSQL**. Since our application deals with tightly linked corporate tables, a **Relational Database Management System (RDBMS)** is strictly necessary to maintain clean data records.

Below is a direct comparison showing why PostgreSQL was chosen over alternative marketplace options:

Feature	PostgreSQL (Selected)	MySQL (Not Chosen)	Oracle Database (Not Chosen)
Price	Free (Open-source)	Free (Open-source basic version)	Extremely Expensive (High license fees)
Speed	Excellent for complex data calculations	Very fast for simple website queries	Unmatched speed for giant bank data volumes
JSON Support	Excellent (Handles messy gateway data easily)	Basic support	Good support but hard to configure
Cloud Fit	Perfectly fits Amazon AWS RDS setups	Fits cloud environments well	Requires special enterprise cloud contracts
Security	High enterprise security features	Good standard security	Maximum security protocols

Final Choice Justification

- **Perfect Pricing:** PostgreSQL provides high-end corporate features without the massive budget expenses of Oracle.
- **Smart Data Handling:** It reads text-based JSON code formats natively. This is critical for our **Integration Gateway** when translating messages from WhatsApp or Odoo.
- **AWS Readiness:** It connects perfectly with the **Amazon AWS Saudi Region**, ensuring our business databases remain highly secure, legal, and fast for local users.

1.6 Non-Functional Requirements (NFRs) & Implementation

1- Performance (السرعة وكفاءة النظام)

- **Requirement (Testable):** The system must process an uploaded Excel sheet containing up to 500 spare part rows and return the initial automated price matches within **5 seconds** max.

- **How Achieved:** We achieve this by using **Node.js** to handle the file upload processing asynchronously in the background. This ensures the client's screen never freezes or locks up while waiting for the calculation to finish.

2- Availability (التوافق والعمل بدون انقطاع)

- **Requirement (Testable):** The entire B2B platform must maintain a **99.9% uptime** record over any given calendar year (allowing less than 9 hours of total downtime for maintenance).
- **How Achieved:** We achieve this by hosting the platform on **Amazon Web Services (AWS)** infrastructure. AWS uses multi-zone server clustering, meaning if one physical data server fails, a backup server automatically takes its place instantly.

3- Data Sync Latency (سرعة تحديث البيانات)

- **Requirement (Testable):** When a local supplier modifies a part price or quantity, the update must sync across the entire platform network and show on the client's screen within **2 seconds (100% success rate)**.
- **How Achieved:** We achieve this using our **Pub/Sub event-driven bus (AWS SNS/SQS)**. Instead of forcing the client portal to constantly ask the database for changes, the supplier portal broadcasts a StockUpdated event that updates client screens immediately.

4- Security & Access Control (الحماية وصلاحيات الدخول)

- **Requirement (Testable):** The platform must reject unauthorized data access requests **100% of the time**. It must block any user from viewing administrative pages if their account role does not explicitly match the Role-Based Access Control (**RBAC**) settings.
- **How Achieved:** We achieve this by creating a middleware validation wall in our **Node.js API Controller**. Every single page request checks the user's secure session token and drops the connection instantly if the account role lacks authorization.

5- Integration Gateway Security (حماية بوابات الربط الخارجي)

- **Requirement (Testable):** The Integration Gateway must block **100%** of external data connection traffic that does not present a verified, unexpired **API token key** during data exchange requests.
- **How Achieved:** The gateway acts as a server-side gatekeeper. Every time Odoo or the WhatsApp API speaks to our system, our integration layer forces an automated handshake verification check before any data reaches the database core.

6- Scalability (القدرة على التوسع وتحمل الضغط)

- **Requirement (Testable):** The platform database and server infrastructure must handle a **300% sudden increase** in concurrent active users (from 1,000 to 4,000 active sessions) during peak business hours without dropping connections.
- **How Achieved:** We achieve this using **AWS Auto-Scaling**. When system memory usage crosses 70%, the hosting cloud automatically fires up extra server clones to split the traffic load evenly.

7- Reliability & Backup (اعتمادية النظام ونسخ البيانات آلياً)

- **Requirement (Testable):** In the event of a critical system crash, the database recovery tools must restore all primary market transaction data with a **maximum data loss limit of 5 minutes (RPO)** and a recovery time under **15 minutes (RTO)**.
- **How Achieved:** We achieve this by configuring **Amazon RDS automated replication**. The database automatically creates point-in-time snapshot log files every 5 minutes and backs them up to separate virtual server storage zones.

8- Financial Accuracy (دقة البيانات المحاسبية والمالية)

- **Requirement (Testable):** The platform integration gateway must sync bulk wholesale transaction data with the Odoo ERP tool with **100% data matching accuracy** (zero transaction rounding or dropping errors allowed).
- **How Achieved:** We achieve this by wrapping our **Node.js Integration Controller** requests inside transactional database blocks. If the external Odoo API drops out mid-way through a payload trade, our gateway rolls back the event entirely so no half-saved or mismatched bills ever exist.

1.7 Business Rules and Business Requirements

1.7.1 Business Rules (قواعد العمل - BR)

BR-01: Profile Status Constraint (قيود حالة الملف الشخصي)

- New corporate client accounts must remain in a locked, pending state and cannot request pricing quotes or browse inventory until an internal system administrator manually reviews and approves their profile.

BR-02: Lowest Price Selection (اختيار السعر الأقل)

- When a client uploads a parts request spreadsheet, the system must automatically analyze and select only the single lowest price available for each specific part number across all registered local suppliers.

BR-03: Order Lifecycle Automation Trigger (تفعيل أتمتة دورة الطلب)

- The exact moment a client officially accepts an aggregated PDF quote, the system must instantly issue automated digital alerts to every individual supplier whose items were selected.

BR-04: Upfront Payment Obligation (الزامية الدفع المسبق)

- Final purchase orders and shipping notifications cannot be released to vendors until the integrated payment gateway successfully confirms 100% collection of funds from the client.

1.7.2 - Business Requirements (أهداف ومتطلبات العمل)

Financial Benefits (الأهداف المالية)

- **Save Money on Parts:** To help wholesale buyers lower their spending on spare parts by **25%** by cutting out middleman brokers and finding the cheapest deals automatically.
- **Make Money from Ads:** To create an extra way to make profit by letting car parts factories buy ad space and put up banners on our website to target repair shops.
- **Get Paid Faster:** To speed up the whole buying process so deals take minutes instead of days, which keeps cash moving quickly through the business.

Non-Financial Benefits (الأهداف غير المالية)

- **Clear Market Pricing:** To give buyers an open and honest view of the lowest available market prices so they know they are getting a fair deal.
- **Help Small Suppliers:** To give small local car part shops an easy digital way to reach big wholesale buyers without needing to spend money on sales teams.
- **Fewer Typing Mistakes:** To stop human data entry errors by making sure all order updates and records are saved and moved automatically between systems.

1.8 System Security Standards (تحديد معايير الأمان للنظام)

1. Presentation Layer Security (حماية واجهات المستخدم)

- **The Threat:**
Hackers may try to steal passwords or view private data.
- **Our Defense (HTTPS & Bcrypt):**
HTTPS protects data during transfer, while Bcrypt secures passwords before storing them in the database.

2. Application & Controller Layer Security (حماية الأكواد وصلاحيات النظام)

- **The Threat:**
Users may try to access information outside their permissions.
- **Our Defense (RBAC & JWT):**
The system checks user roles through secure login tokens and only allows approved access.

3. Integration Gateway Security (حماية بوابات الربط مع الأنظمة الخارجية)

- **The Threat:**
External systems may send fake or unsafe requests.
- **Our Defense (API Verification):**
The gateway accepts only verified requests using secure API tokens.

4. Event Bus Security (حماية ناقل الأحداث والرسائل)

- **The Threat:**
Background messages may be accessed or changed.
- **Our Defense (Private Cloud Network):**
Event communication is kept inside a private network with limited access.

5. Infrastructure & Database Protection (حماية البنية التحتية وقاعدة البيانات)

- **The Threat:**
Attackers may overload the system or target the database.
- **Our Defense (AWS Shield & Private Database):**
AWS filters harmful traffic, while the database stays private and accessible only to trusted services.

1.9 Functional requirement scenario description

Client Portal

Functional Requirement (FR)	Human-Like Real-Life Scenario / Description
The system allows the user to create a new corporate account profile.	Ahmed runs a massive automotive fleet repair workshop in Riyadh. He visits PartZone for the first time, clicks 'Register Workshop,' and types in his workshop's commercial registration details, official name, and contact info to create his primary business account.
The system allows the user to log into the portal.	Once his account gets approved, Ahmed opens the website on Sunday morning, enters his registered email address along with his secure password, and hits the login button to enter his personal dashboard.

The system allows the user to manage account profile information.	Ahmed's workshop relocates to a larger facility down the highway, so he opens his profile settings tab, modifies his business location phone lines, and saves the updated contact details.
The system allows the user to delete their account profile.	If Ahmed decides to retire or sell his workshop operations entirely, he can click on the security settings panel and select 'Close Business Account' to permanently remove his corporate info from the active client base.
The system allows the user to upload an Excel sheet to request automatic part prices.	Ahmed has 15 cars waiting for repairs. He opens a blank Excel sheet, types a column of 45 different manufacturer spare part numbers along with their requested quantities, and drags that file directly into the PartZone upload box to look for matching bids.
The system allows the user to use integrated AI tools to read and compare different pricing options.	Instead of opening dozens of vendor quotes manually, Ahmed clicks 'Run AI Price Optimizer.' The platform's internal engine automatically reads every available vendor's inventory file and filters out the absolute cheapest matches for each separate row on his sheet.
The system allows the user to view and download a generated PDF quotation sheet.	After the AI completes its filtering magic, a pop-up button appears saying 'Download Combined Quote.' Ahmed clicks it, and an elegant PDF file opens up showcasing a neat breakdown of the cheapest market rates along with a final grand total.
The system allows the user to accept or reject the offer.	Ahmed reviews the total cost on the PDF, finds it incredibly affordable, and hits the big green 'Accept and Purchase' button on his screen to instantly push the order into the fulfillment stage.
The system allows the user to review new registration requests.	(Admin View) The system admin logs into the gateway and checks a dedicated pending verification table to see a fresh application submitted by Ahmed's workshop.
The system allows the user to approve a new client account to give them system access.	(Admin View) The system admin verifies Ahmed's corporate license documents, clicks the 'Verify Account' checkmark, which instantly changes Ahmed's system status from pending to active and sends him an invite.

Admin Dashboard

Functional Requirement (FR)	Human-Like Real-Life Scenario / Description
The system allows the user to create new internal employee or manager accounts.	The marketplace platform expands its team, so the main system Admin opens the staff control tab, inputs a new employee's corporate details, and creates their official credentials to let them log into the back office.

The system allows the user to delete internal employee or manager accounts.	When an office manager resigns from the company, the Admin opens the master staff list directory, clicks 'Revoke Account Permissions,' and deletes that user's login access permanently to protect platform security.
The system allows the user to assign roles and permissions to internal staff members.	The Admin wants to make sure data remains locked, so he modifies an employee's account privileges using an RBAC dropdown menu, toggling their access level so they can view active logistics paths but cannot touch core system configurations.
The system allows the user to review and approve pending wholesale client profiles.	The Admin opens his dashboard queue first thing in the morning, reads through a list of newly uploaded corporate registration records, checks their validity, and clicks approve to grant those businesses bidding capabilities.
The system allows the user to monitor active order lifecycles.	A combined manager/employee named Khalid pulls up the system's live workflow monitoring screen to track Order #7704, checking whether it has successfully transitioned from 'Paid' to 'Dispatched' at the supplier's warehouse.
The system allows the user to track general platform processes and system workflows.	Khalid monitors the background system metrics graphs to verify that the automated AI price matching operations are processing uploaded client Excel sheets without any performance lag or connection drops.
The system allows the user to publish manual system updates to the event bus.	During an emergency system maintenance window, Khalid manually inputs an alert message into an admin panel console and hits broadcast, publishing a system event notification to gracefully notify all active supplier portals.
The system allows the user to flag client profiles for administrative review.	While auditing active transactions, Khalid notices a user account uploading highly suspicious spreadsheet payloads, so he clicks the warning icon next to their name to lock the profile and flag it for full review.

Integration Gateway (System-to-System Connections)

Functional Requirement (FR)	Human-Like Real-Life Scenario / Description
The system allows the external system to verify secure connection keys before trading any data.	The external Odoo accounting server tries to push a transaction log into PartZone. Before letting it past the digital firewall, our gateway interceptor stops the connection and demands an encrypted API secret key token to prove it is authentic.
The system allows the external system to validate all incoming and outgoing data traffic.	When business metrics move back and forth across the web, the gateway monitors the communication pipes to filter out corrupt payloads or malicious tracking code injections before they ever touch our database core.

The system allows the external system to translate data formats before they reach the main application core.	An outside shipping software sends a tracking status update encoded in a raw, custom XML file layout. The gateway instantly grabs this package and rewrites it into a clean JSON data format that our internal platform can naturally understand.
The system allows the external system to convert logistics tracking updates and accounting data into internal system messages.	The Google Maps API or a delivery provider pushes an update saying 'Package departed warehouse.' The gateway catches this webhook and converts it into a native platform update message to alert the user.
The system allows the external system to register as a publisher or subscriber to events on the platform message network.	When the WhatsApp notification gateway connects to our platform, it signs up as a 'Subscriber' to our event hub, waiting silently in the background for any NewOrderConfirmed event to fire so it can trigger an auto-message.
The system allows the external system to pass business data back and forth without connecting directly to internal tools.	When the Odoo ERP tool syncs financial tax logs at midnight, it drops the ledger lines entirely inside the secure gateway bridge, keeping the outside software completely separated from hammering our live transactional database tables.

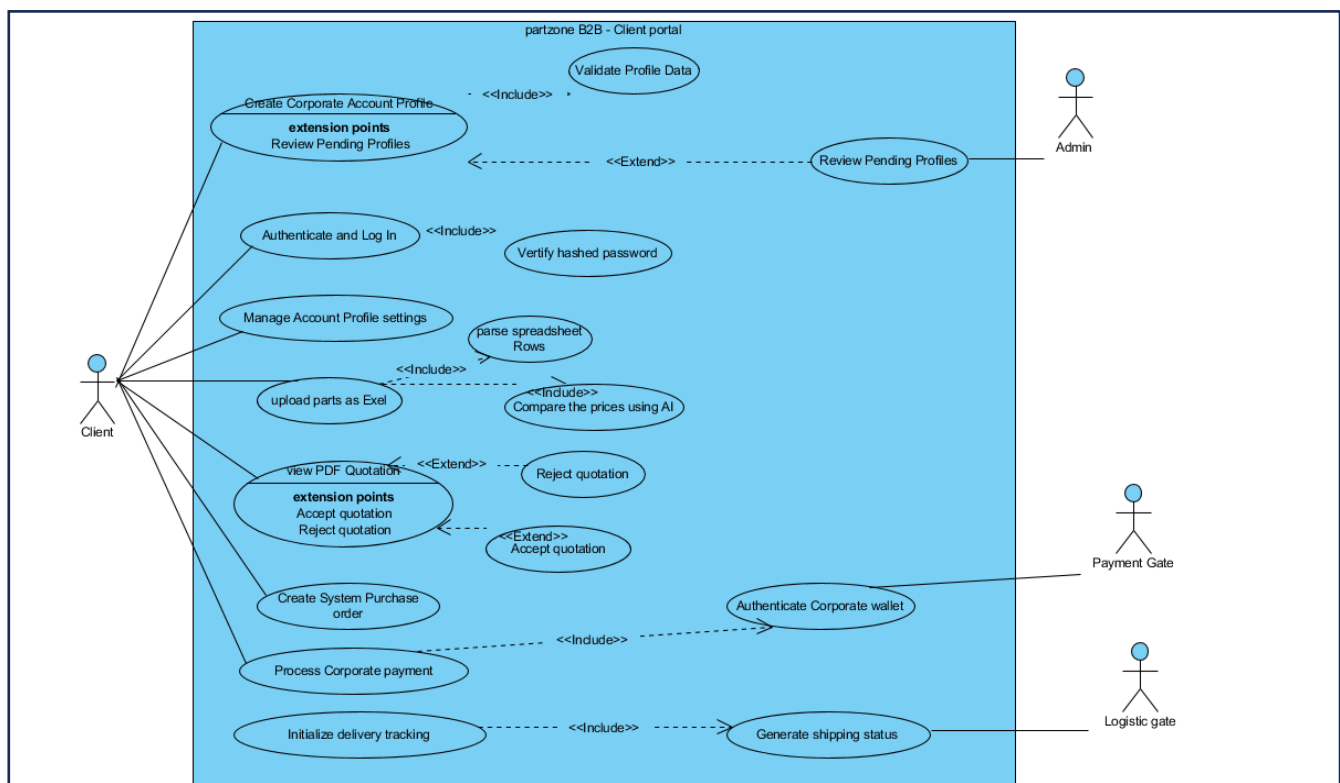
Local Supplier Portal

Functional Requirement (FR)	Human-Like Real-Life Scenario / Description
The system allows the user to create a vendor profile account.	Saeed owns a large automotive wholesale parts warehouse in Jeddah. He visits the platform, registers his corporate merchant store profile, and sets up his vendor identity to start selling stock online.
The system allows the user to manage their corporate account details.	Saeed updates his warehouse's primary contact numbers and business hours within the supplier dashboard settings to ensure logistical dispatch agents can reach his crew smoothly.
The system allows the user to delete their vendor profile account.	If Saeed closes his regional branch warehouse operations, he can navigate to account management, select 'Terminate Supplier Status,' and request to delete his merchant catalog records from the system.
The system allows the user to add new product items to their stock list.	Saeed imports a fresh shipment of 500 ceramic brake pads for Japanese cars. He opens his supplier inventory tab, inputs their factory OEM codes, and logs them into his digital warehouse listing.
The system allows the user to view their personal warehouse product items.	Saeed opens his main portal dashboard page to read a clean, paginated list of all the different spare parts he currently has listed for sale on the platform.

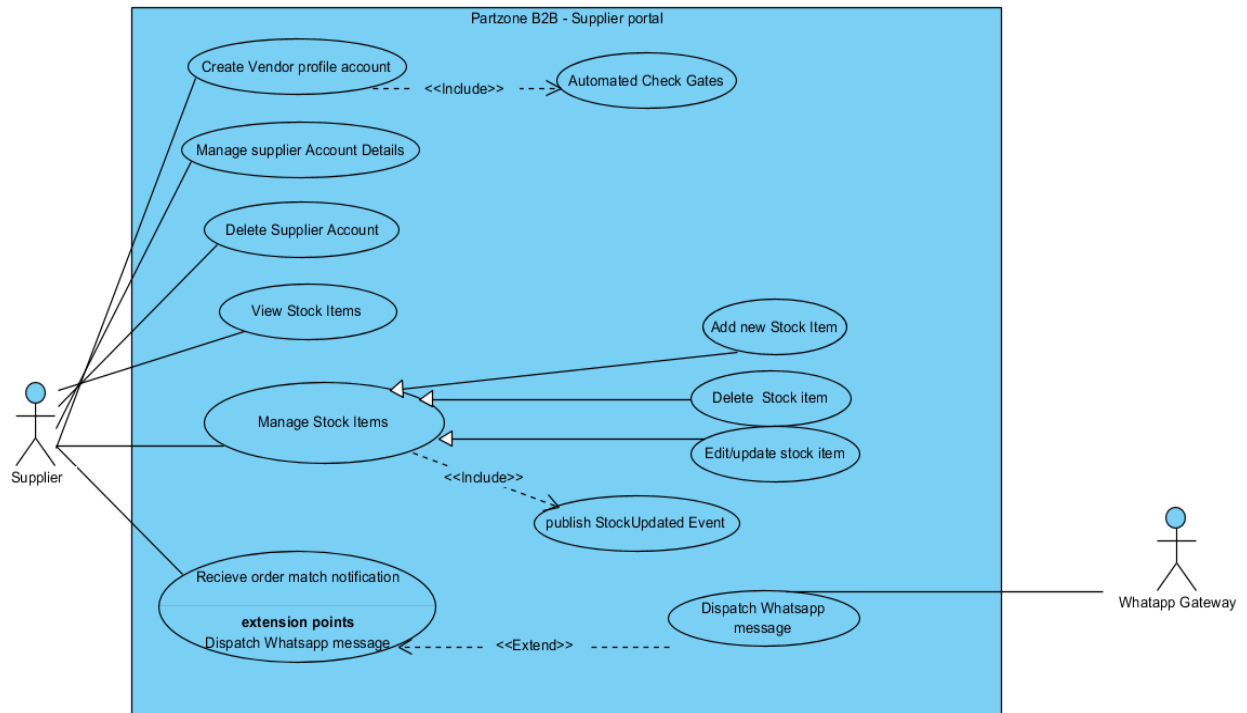
The system allows the user to edit existing product prices and current quantities.	Saeed runs a seasonal clearance promotion, so he opens his stock table, adjusts the unit price of his specific spark plug items down by 10%, updates the quantity counter, and hits save.
The system allows the user to delete items from their online stock list.	When a particular line of older vehicle parts gets permanently discontinued by the manufacturer, Saeed clicks the trash bin icon next to those rows to delete them from his digital store catalog.
The system allows the user to receive automated alerts when a buyer accepts a quote.	The moment Ahmed (the buyer) clicks accept on a PDF offer containing Saeed's brake pads, Saeed's portal instantly rings with a prominent dashboard alert saying 'New Purchase Order Received! Prepare for Dispatch'.
The system requires all typed supplier inputs to pass automated validation check gates on the server side.	Saeed tries to type a negative number or letters into the 'Item Unit Price' input field. The portal backend validation layer immediately catches the bad data format, blocks the submission, and displays a clean error message.
The system allows the user to publish a "StockUpdated" event to the message network when stock numbers change.	When Saeed updates his inventory counts on his screen, the backend code triggers and broadcasts a StockUpdated message to the central event bus, ensuring the client-side pricing calculators see the new quantities with zero latency.

1.10 – Use case for each portal

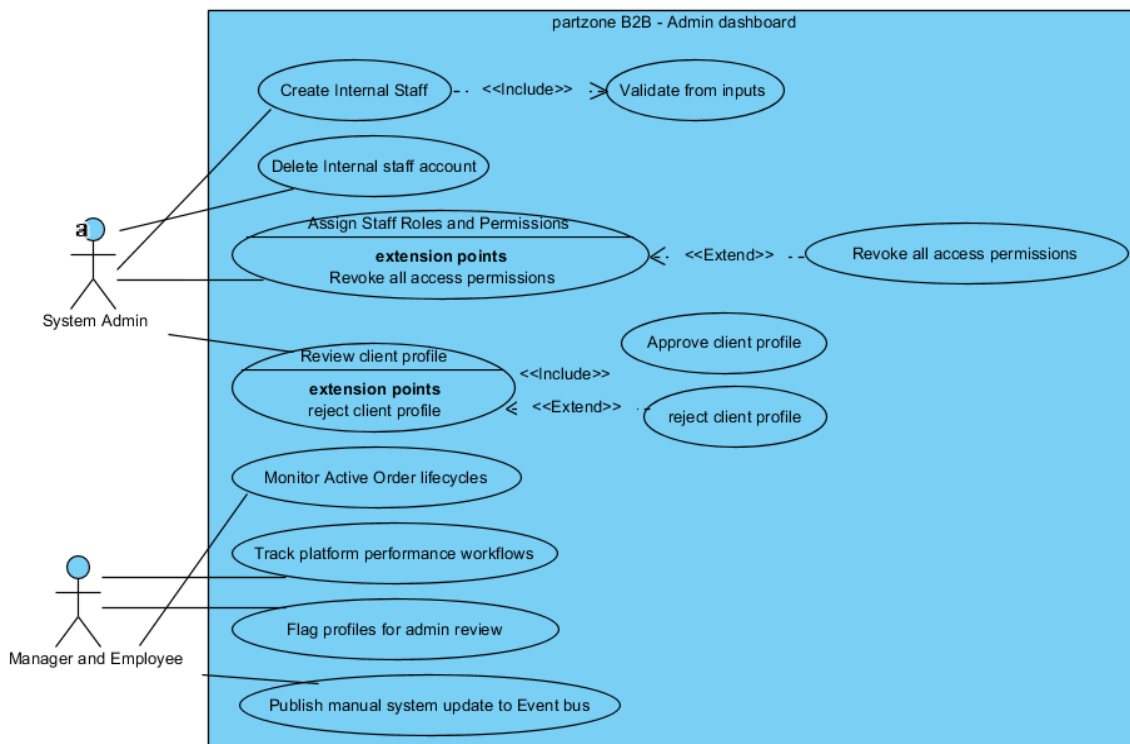
1.10.1 - Client portal use case



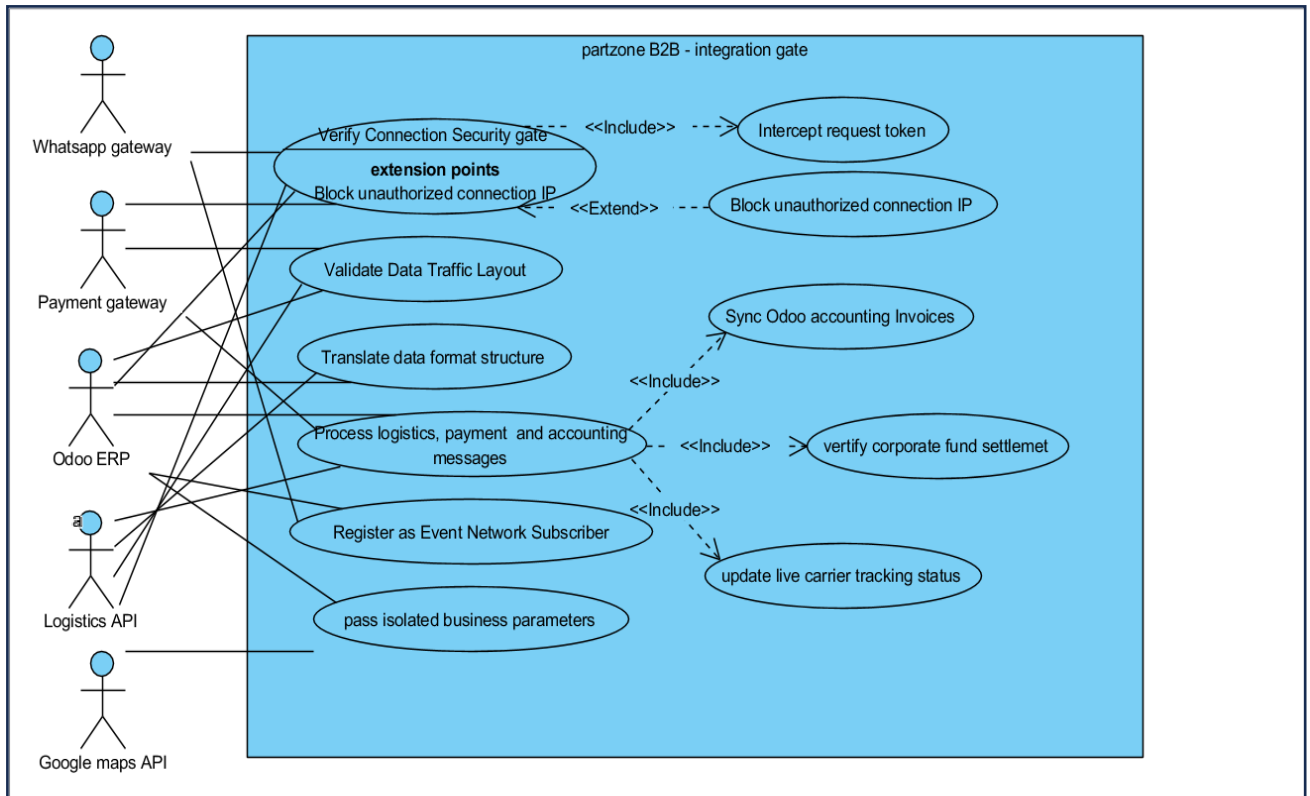
1.10.2 – Supplier portal use case



1.10.3 Admin dashboard use case

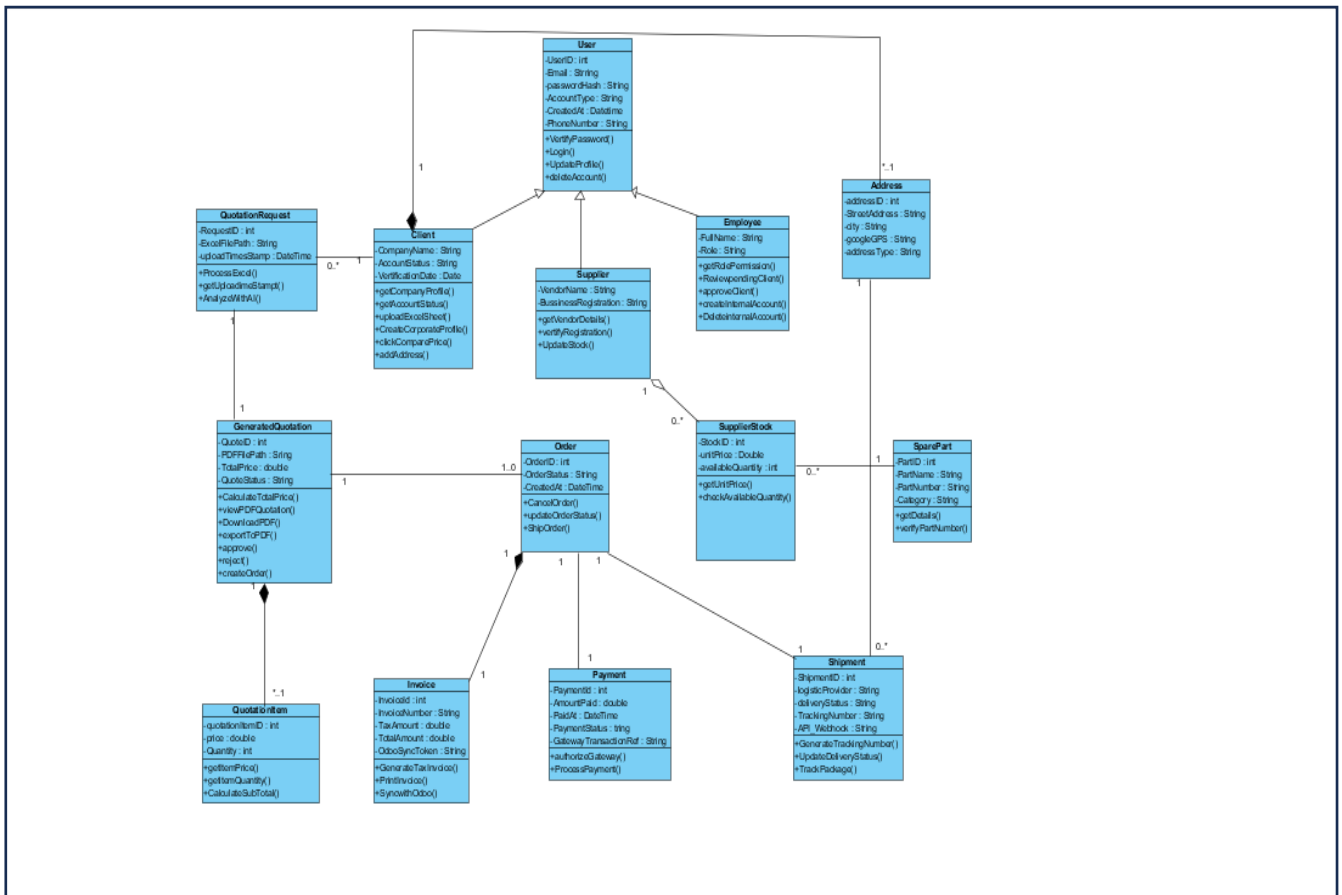


1.10.4 integration gate use case

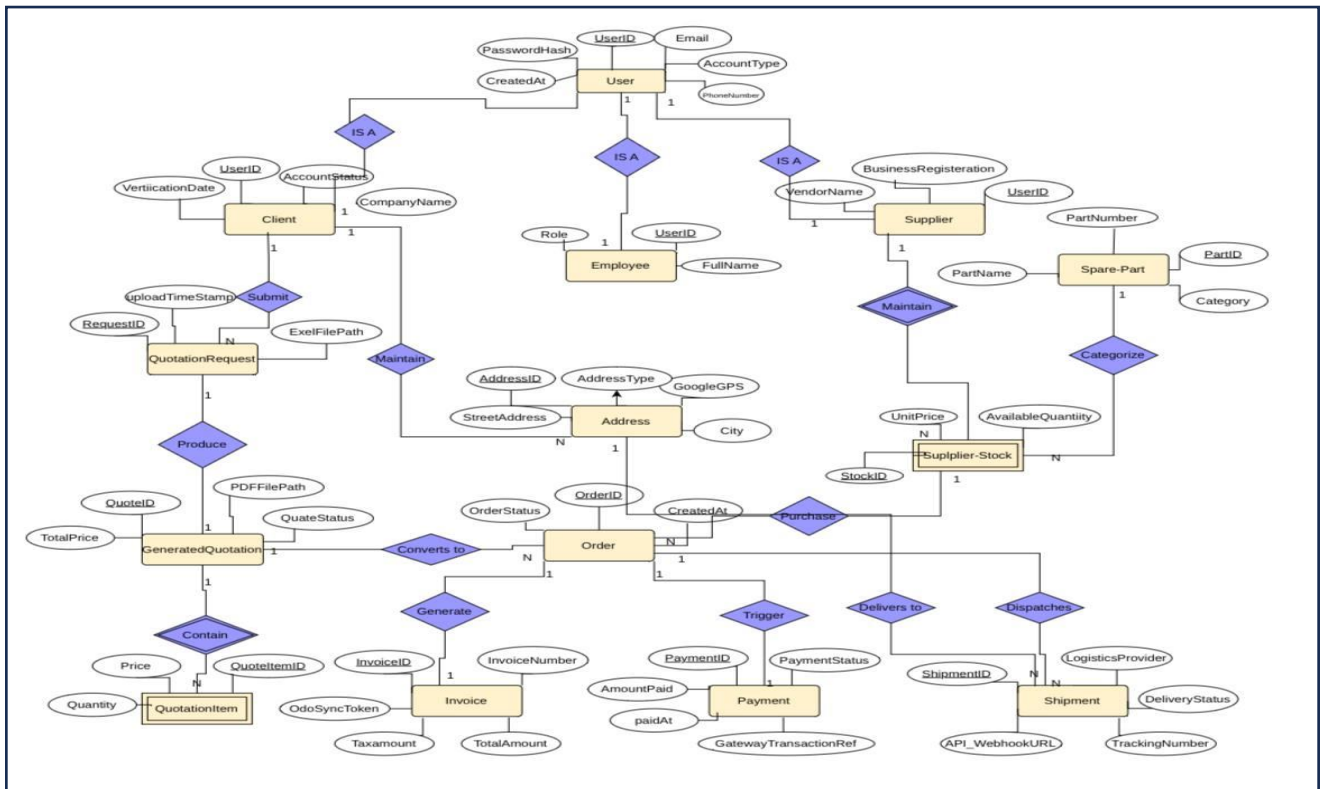


Phase 2: design

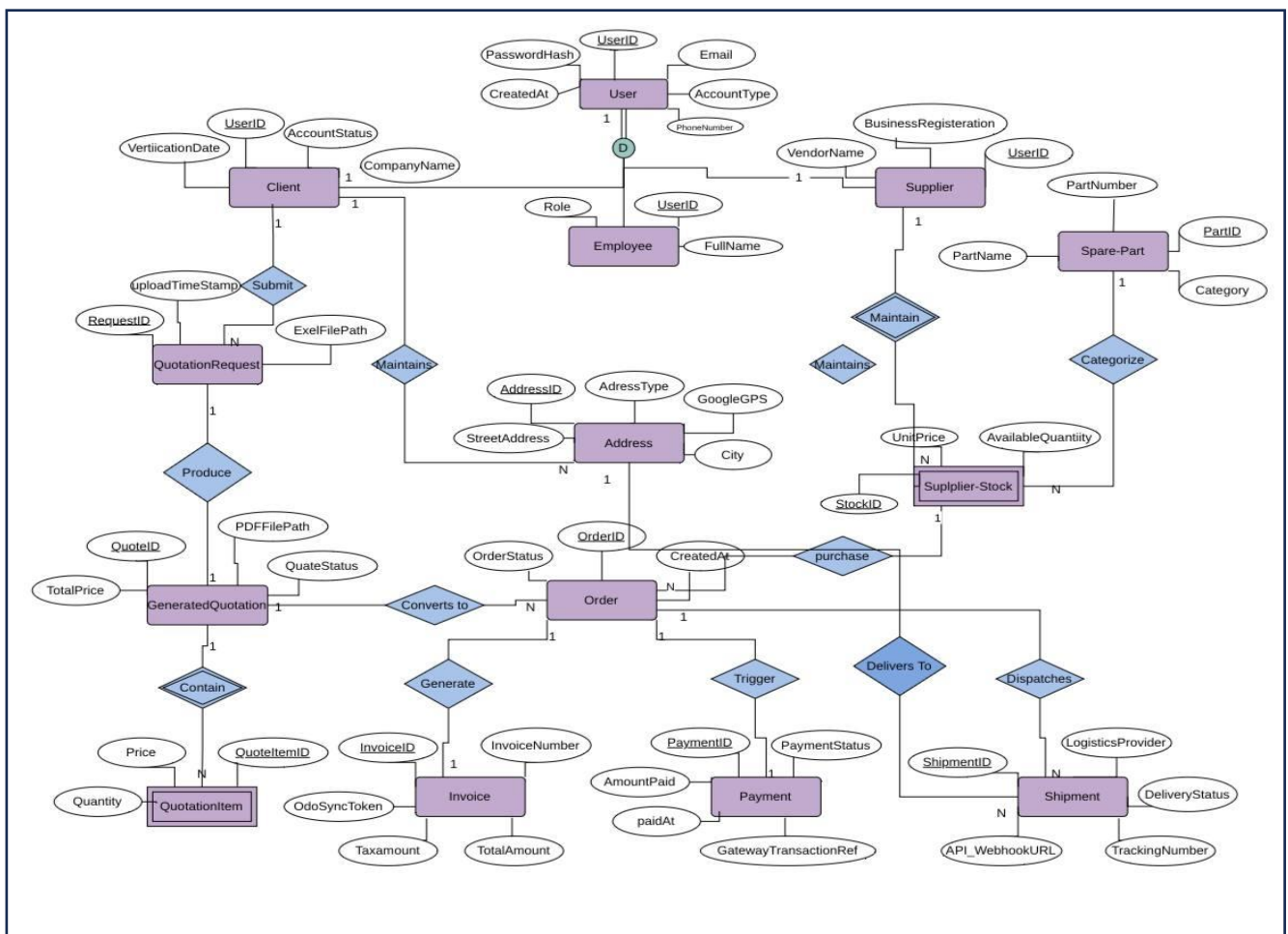
2.1 Class diagram



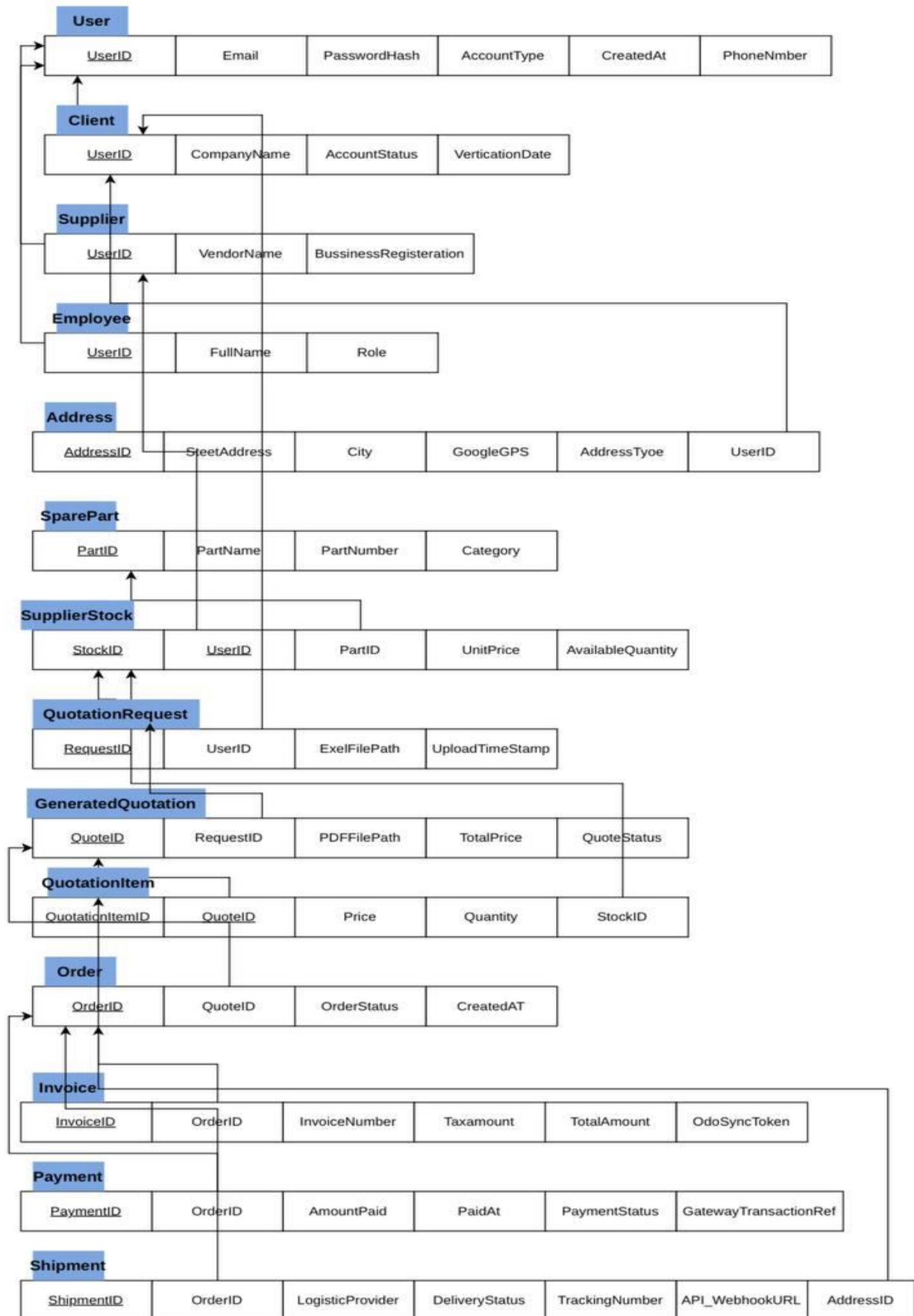
2.2.1 ERD



2.2.2 EERD

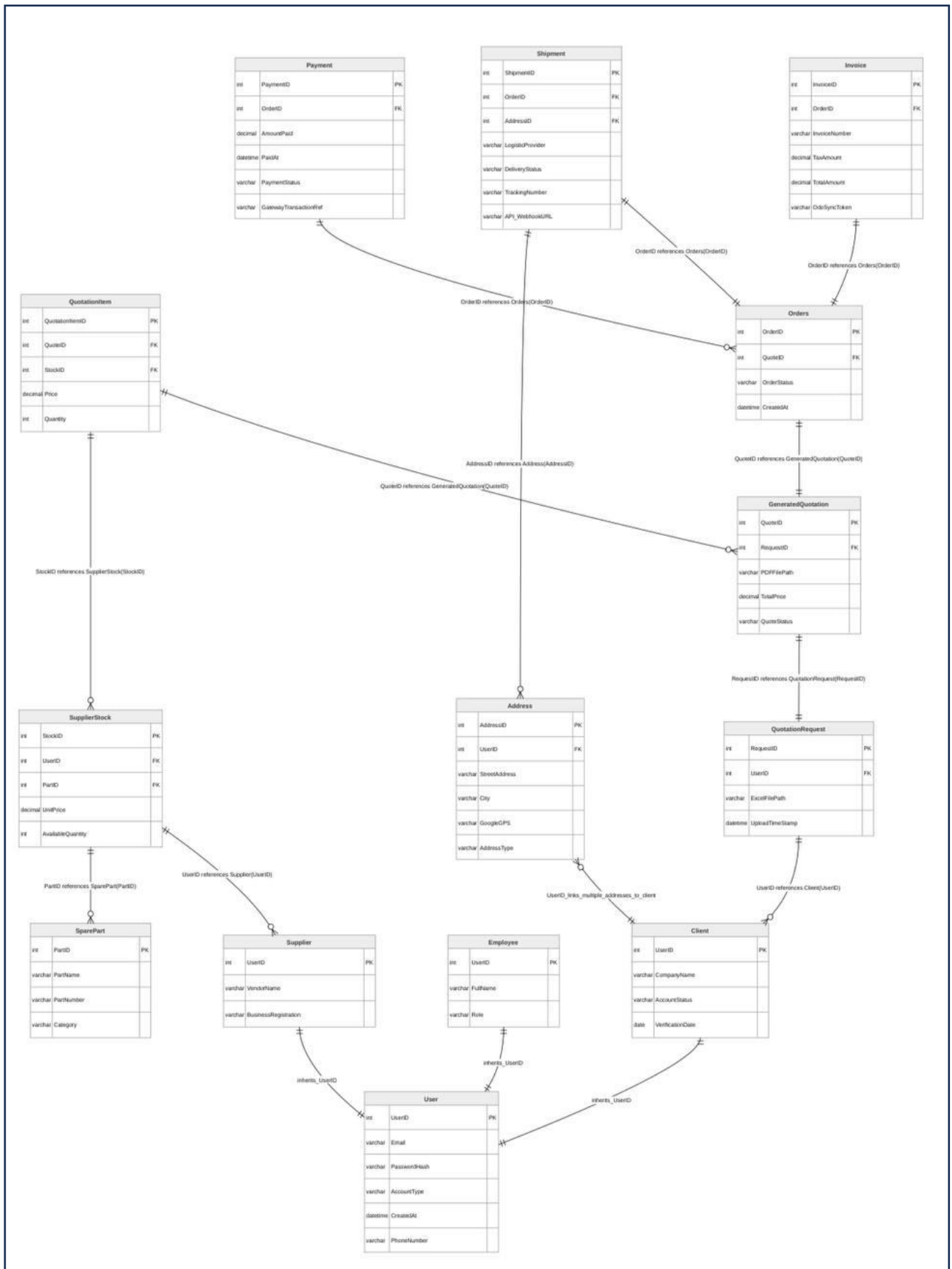


2.2.3 Relational Mapping

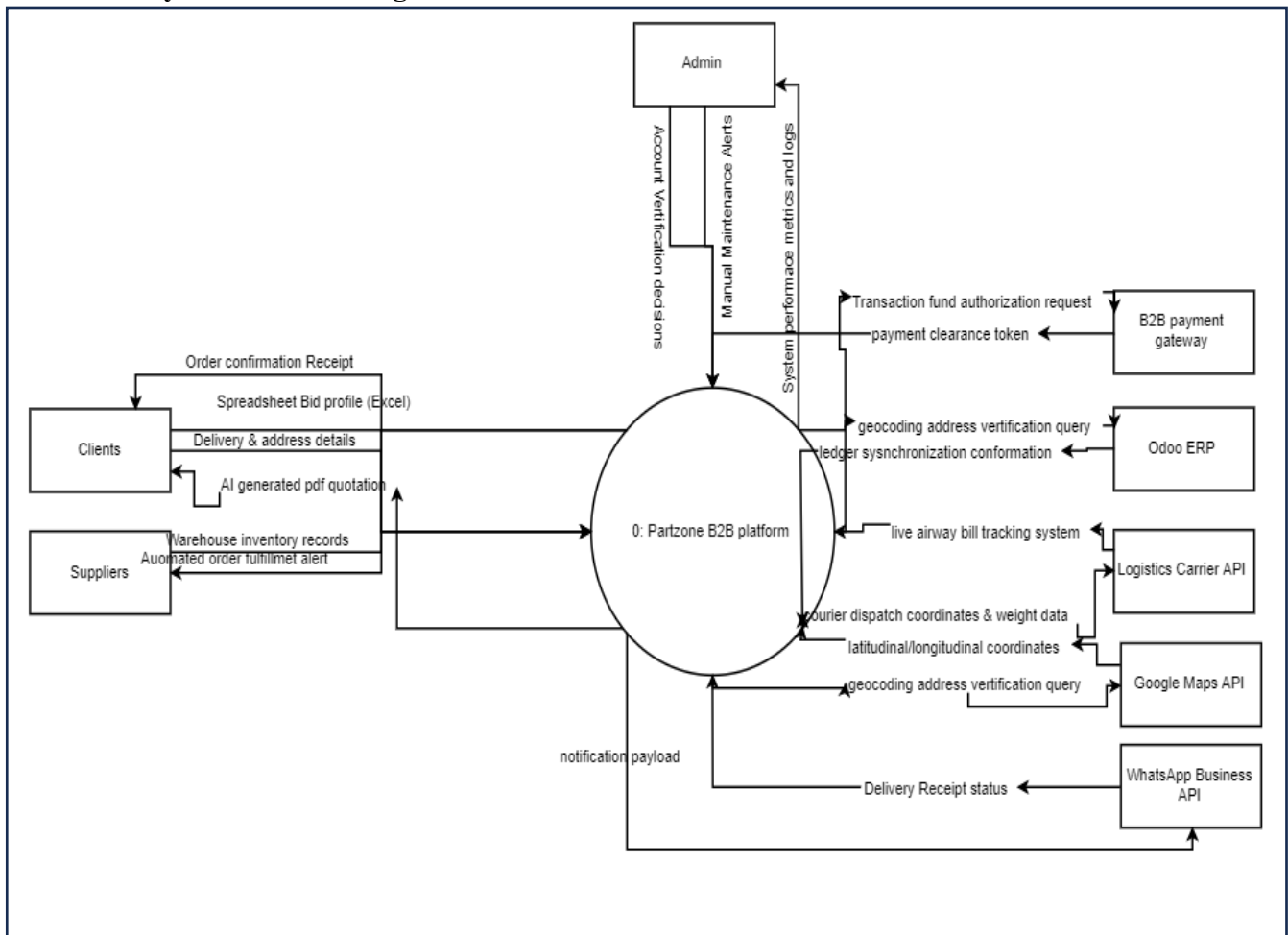


2.2.4 Schema

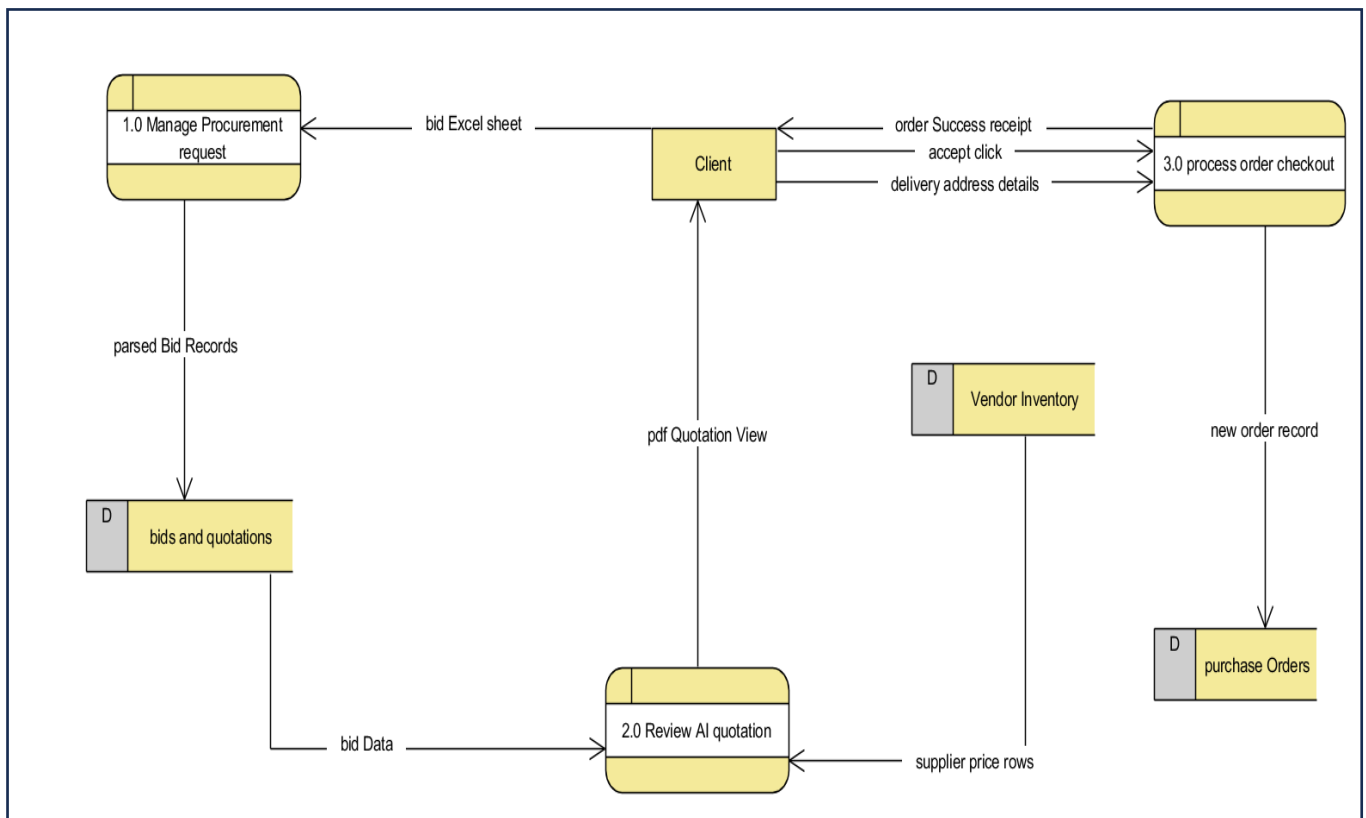
Note: I didn't find it in visual paradigm application, so I did it using draw.io



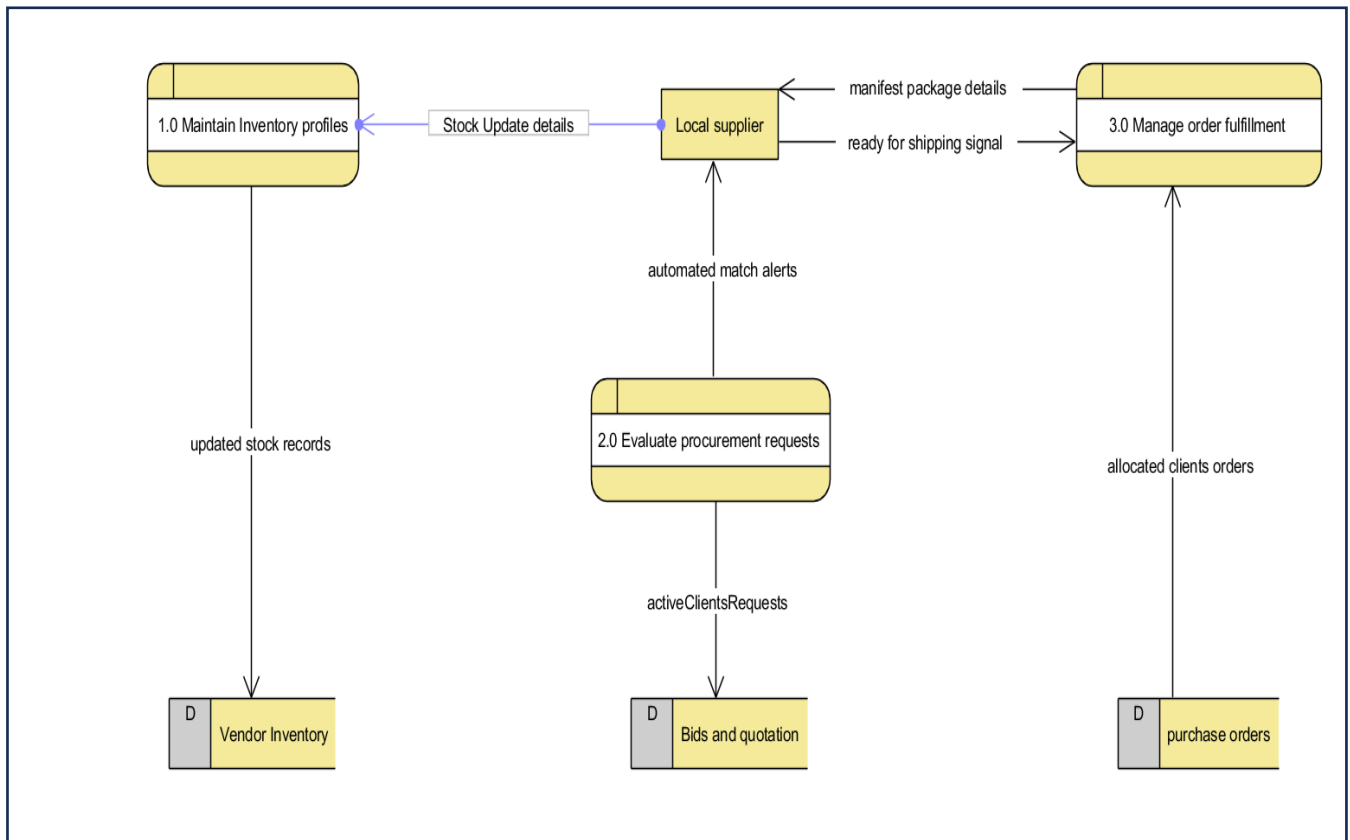
3- System context diagram



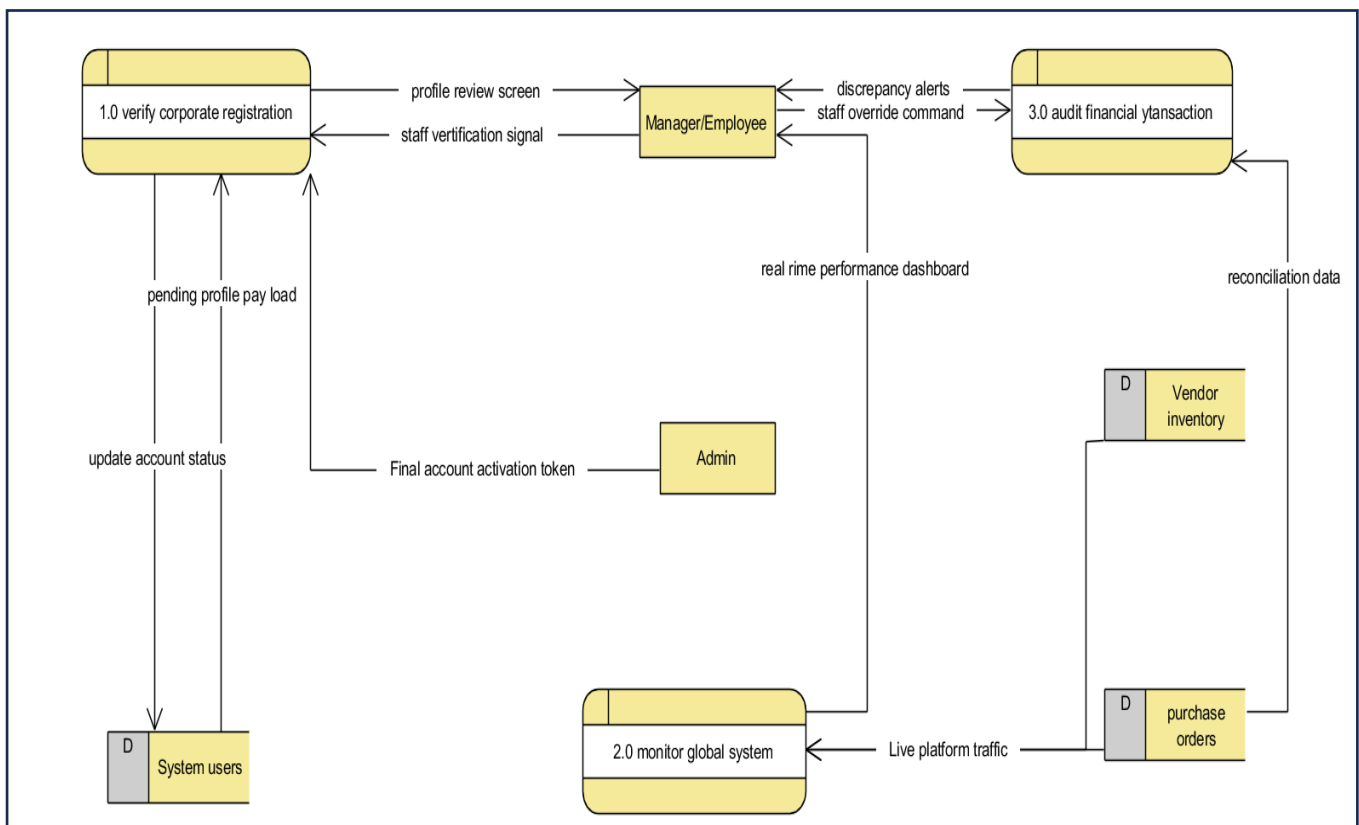
4.1- Data Flow diagram one – client portal



4.2 Data flow diagram one – supplier

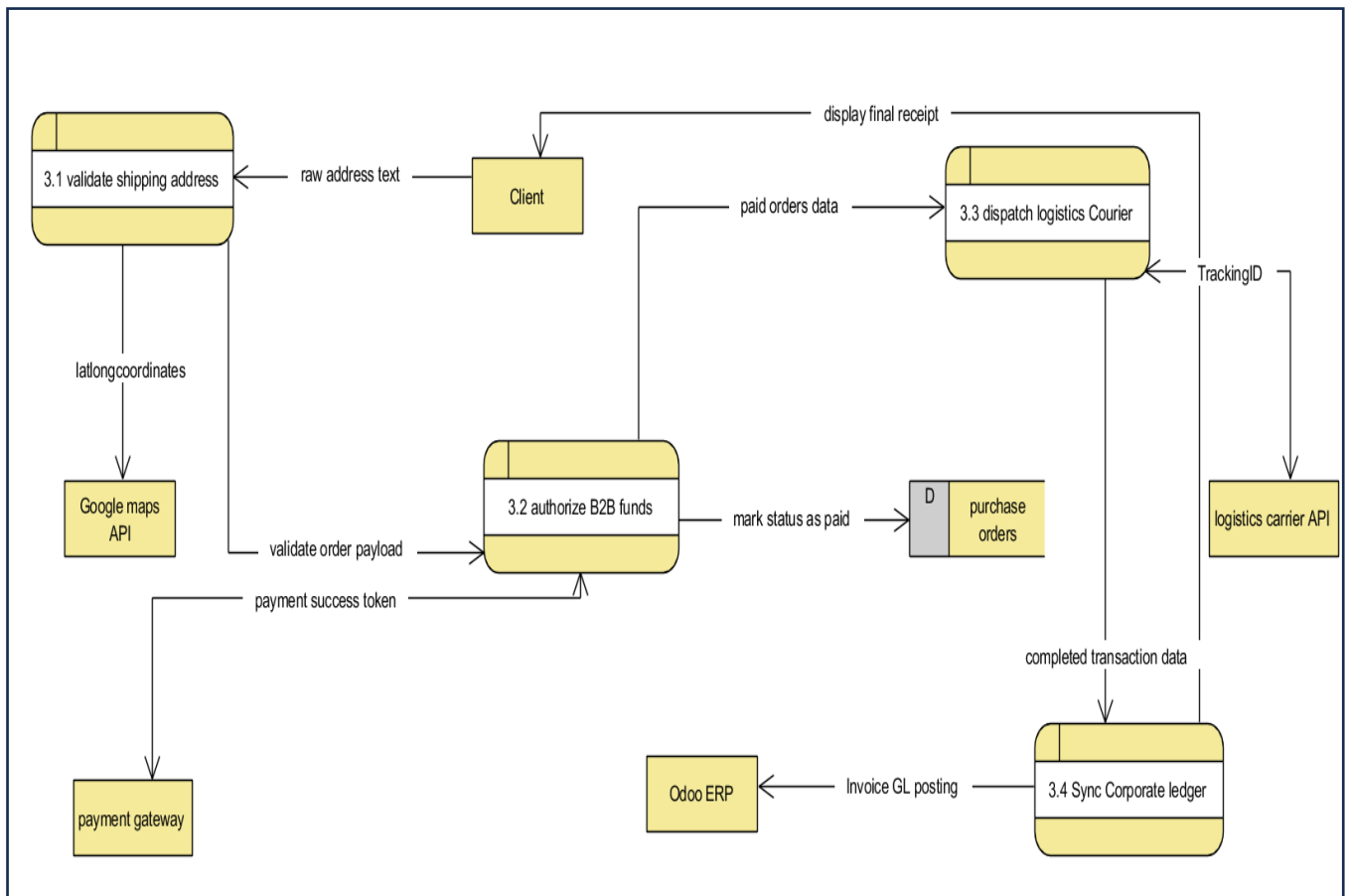


4.3 Data flow diagram one – admin dashboard

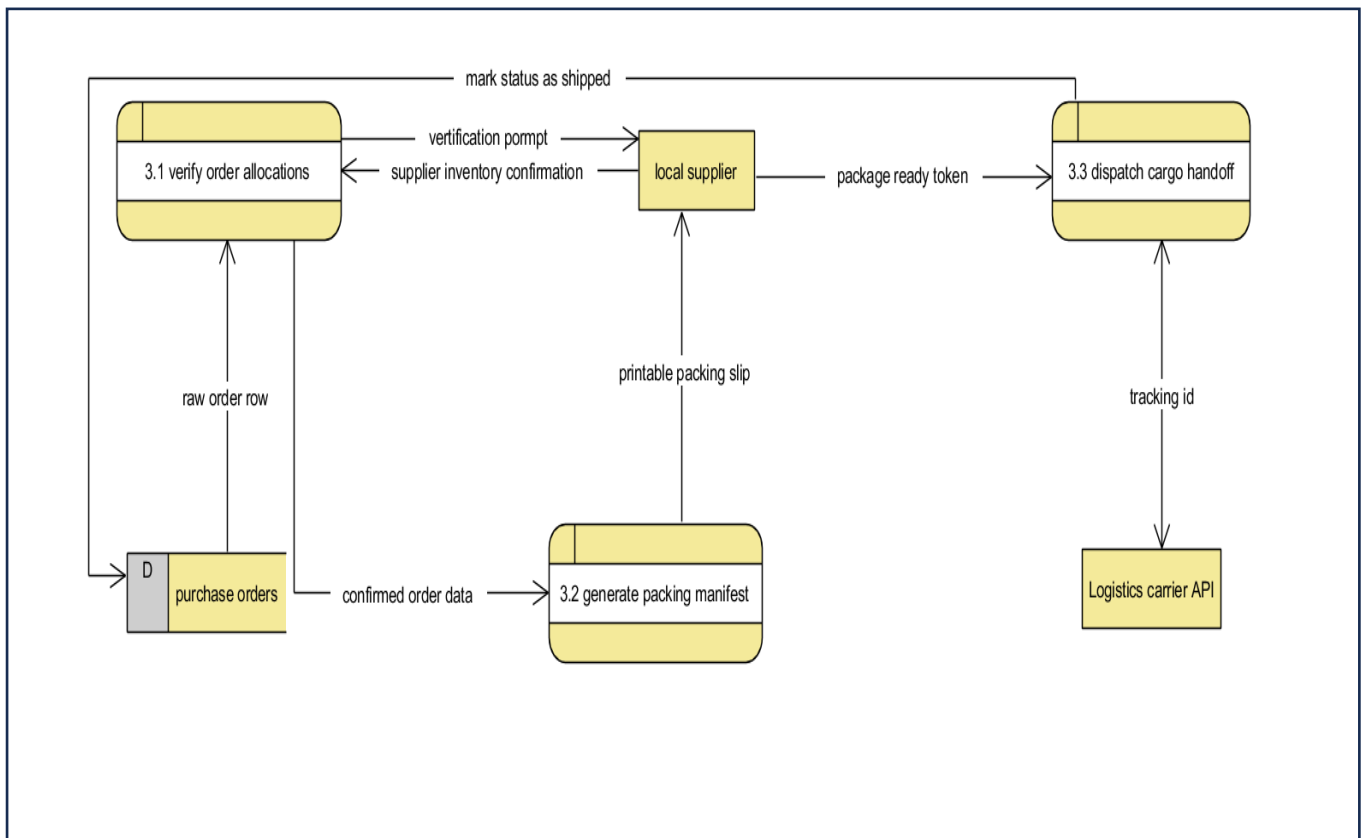


Note: I integrated the external systems inside the main 3 portals DFD, so I didn't do separate one for it

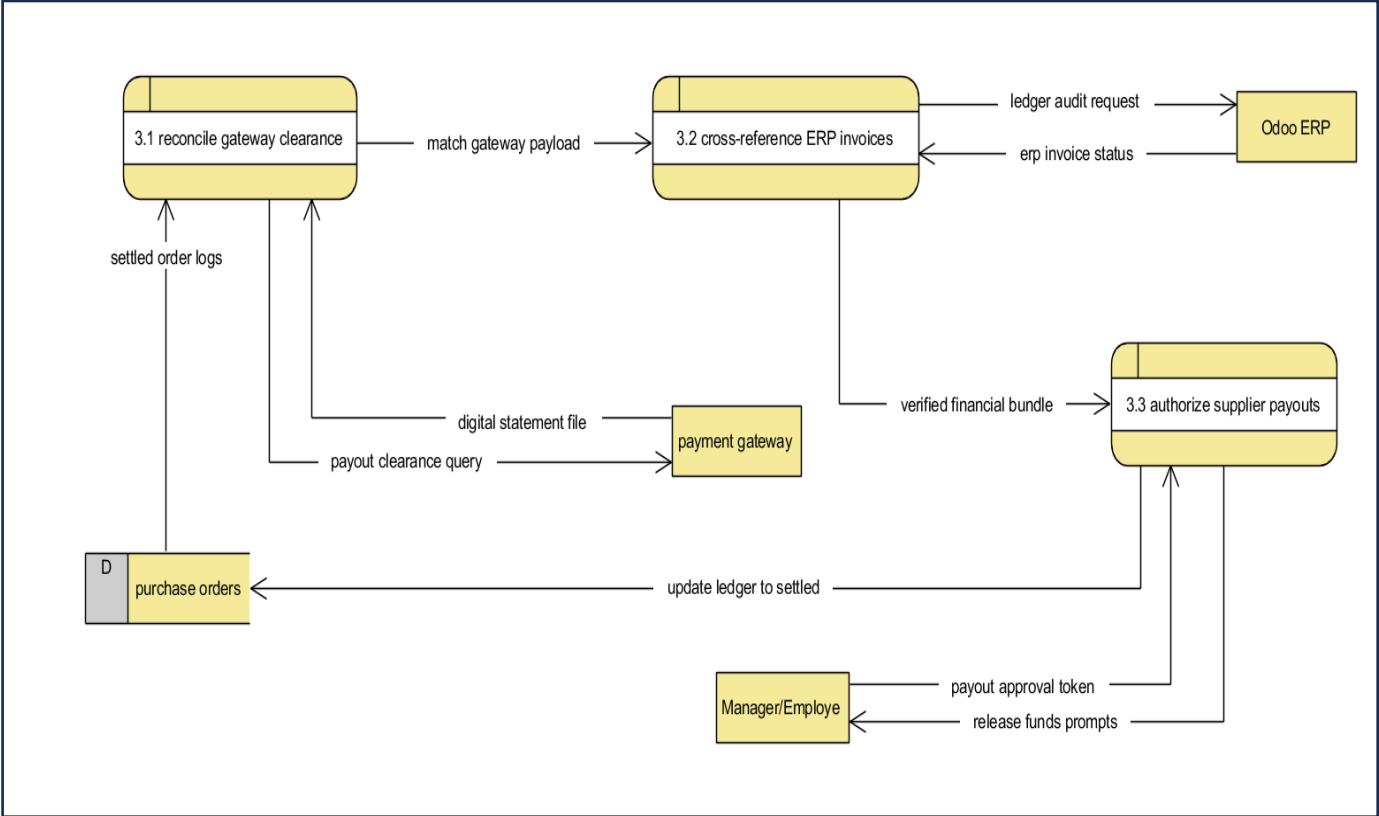
4.4 Data flow diagram two – client portal



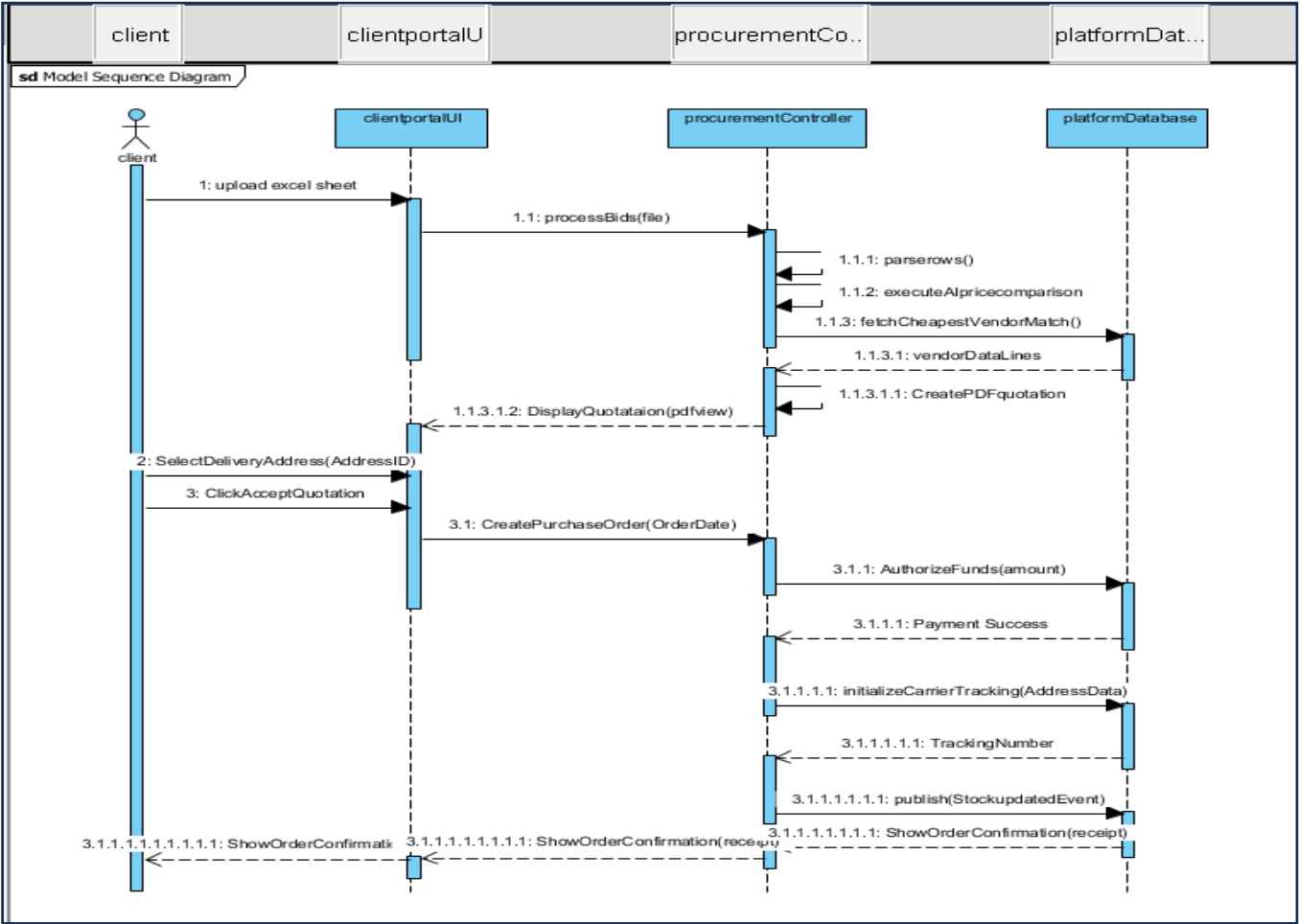
4.5 Data flow diagram two – supplier portal



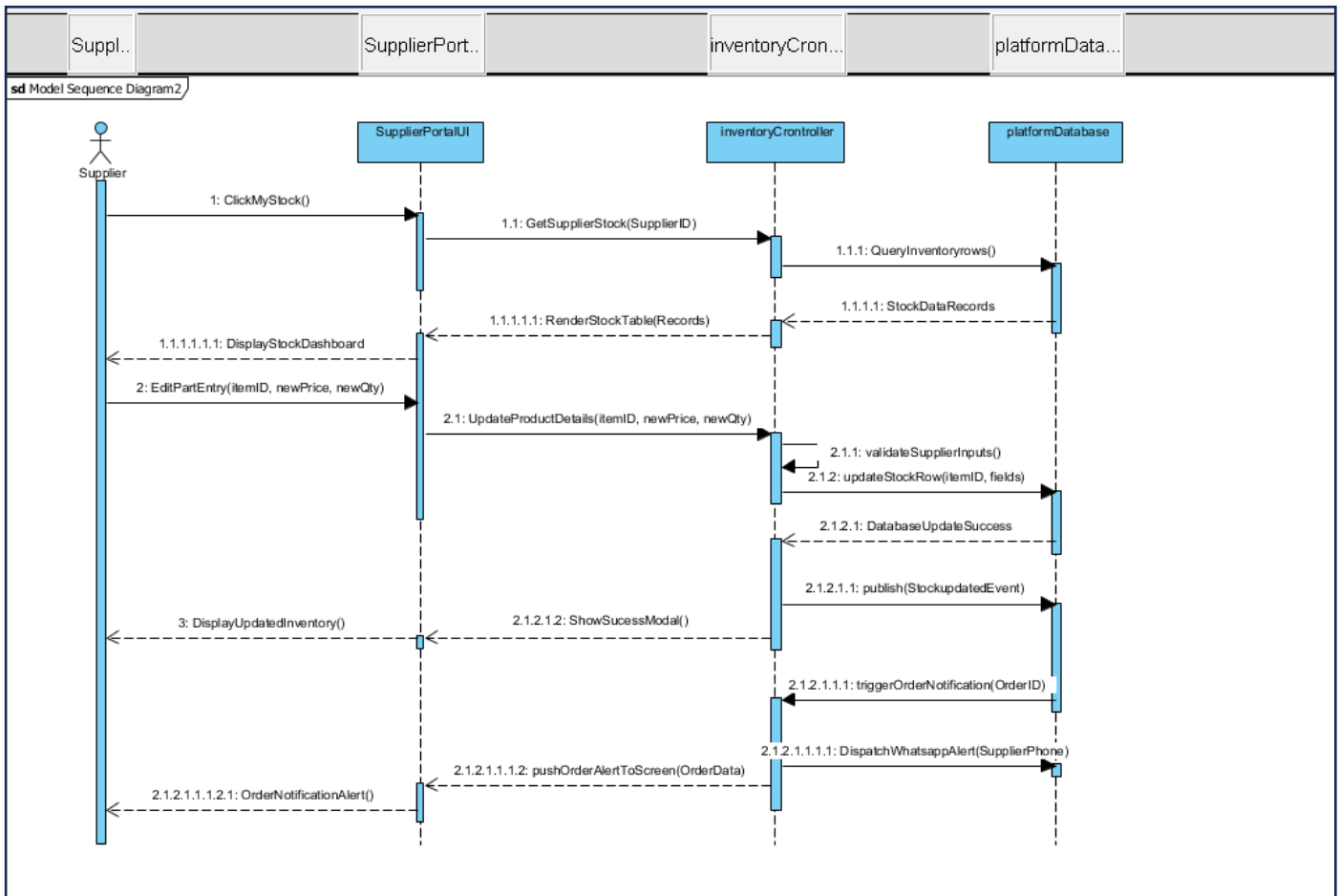
4.6 Data flow diagram two – admin dashboard



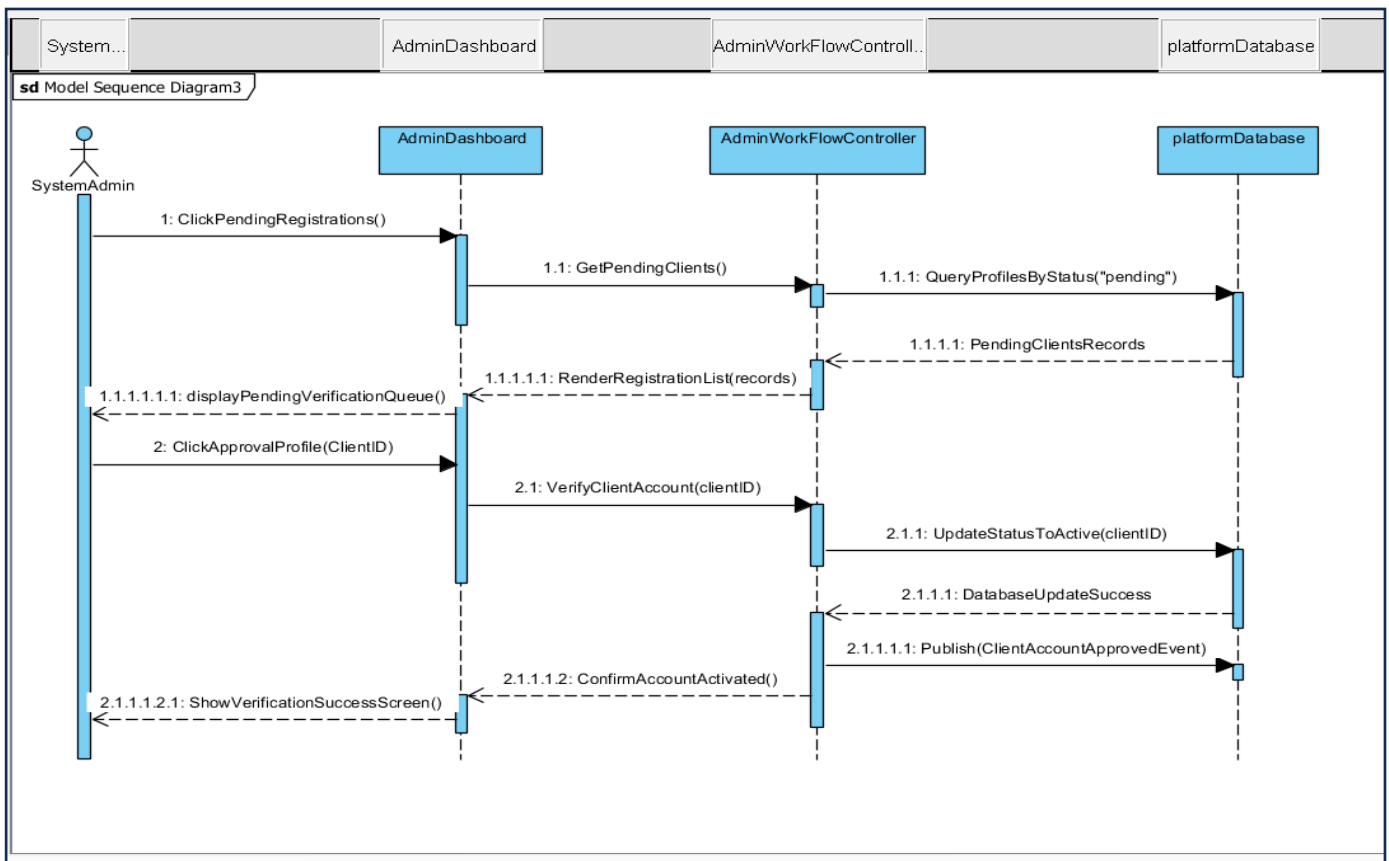
5.1 Sequence diagram – client portal



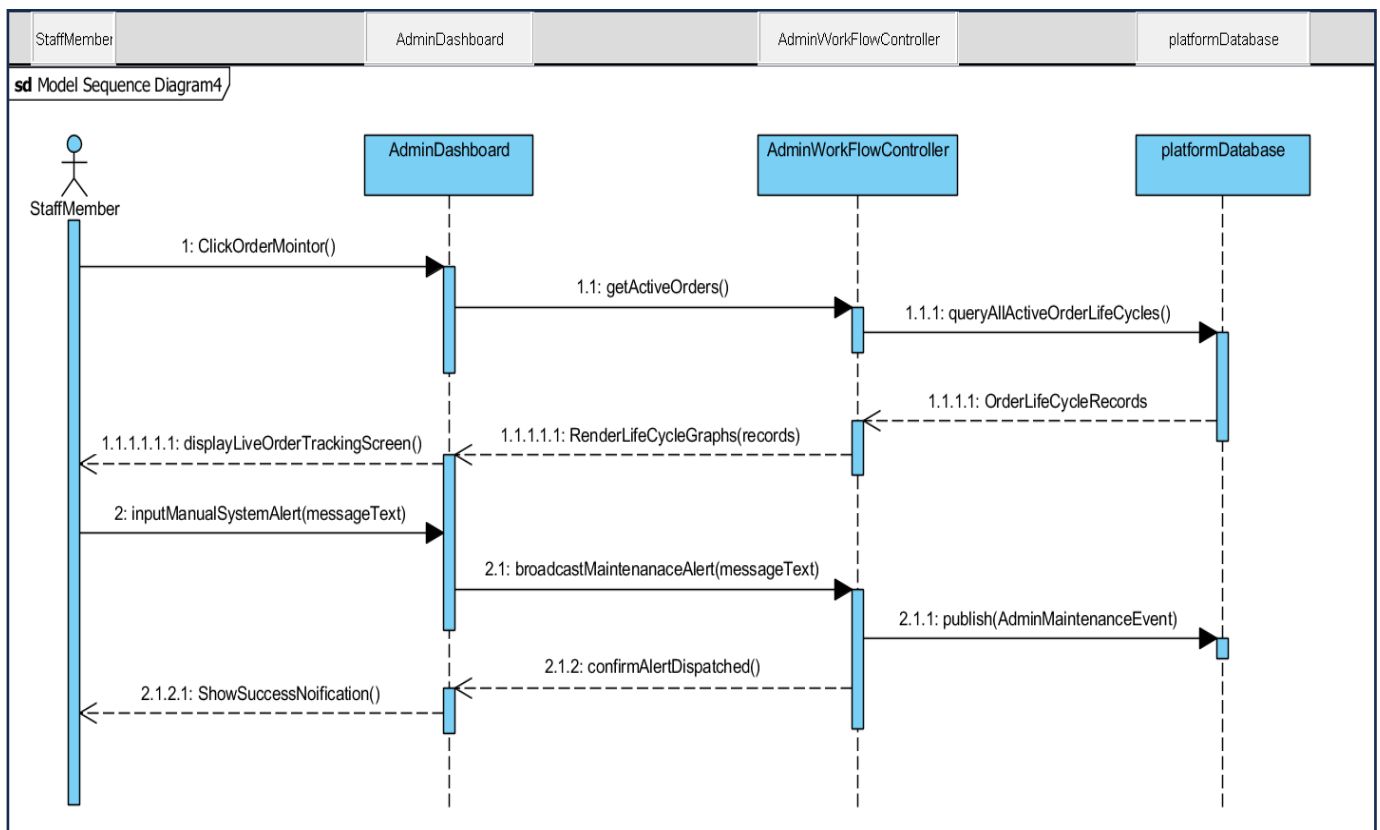
5.2 Sequence diagram – supplier portal



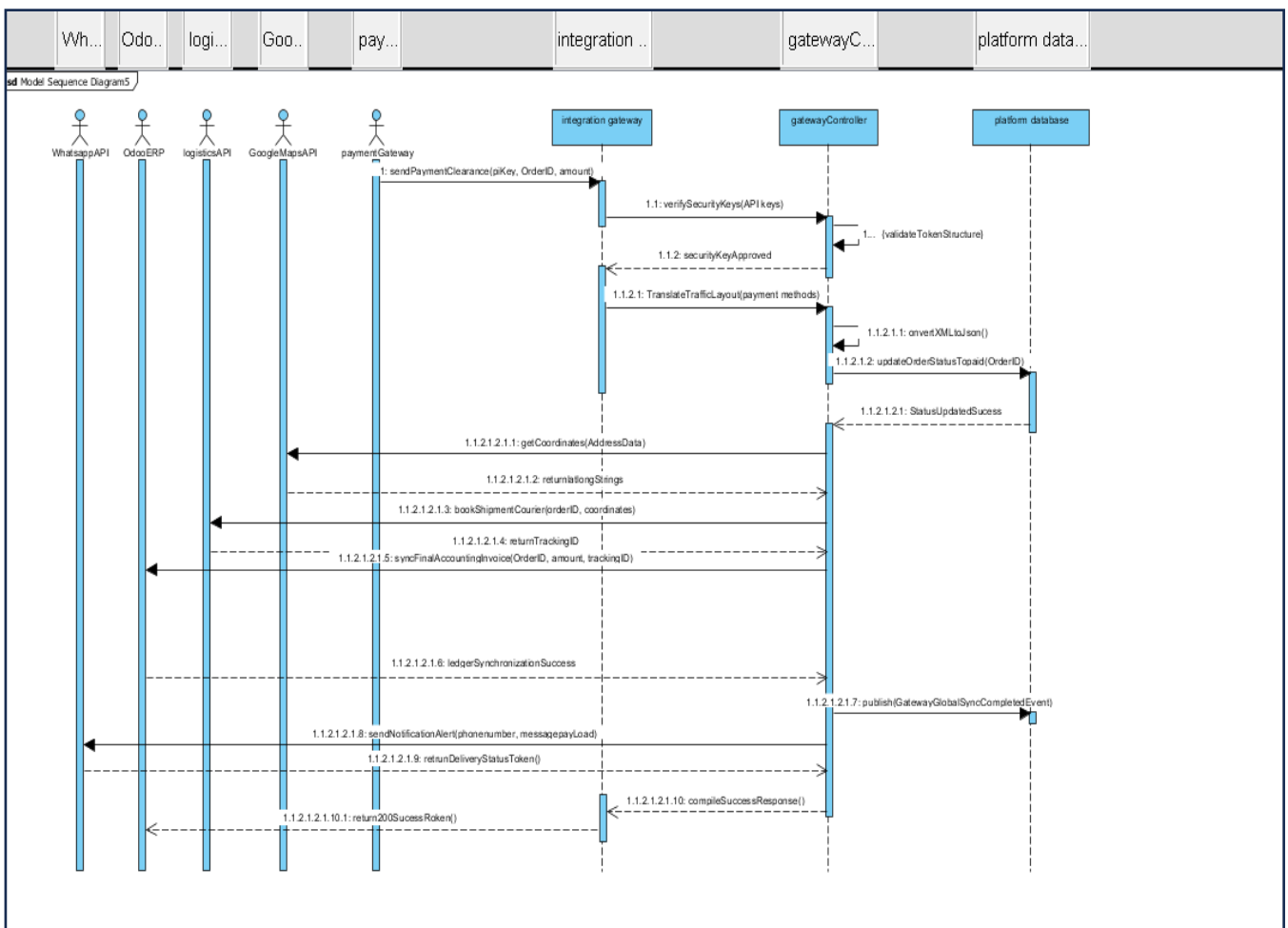
5.3 Sequence diagram – Admin dashboard (Admin actor)



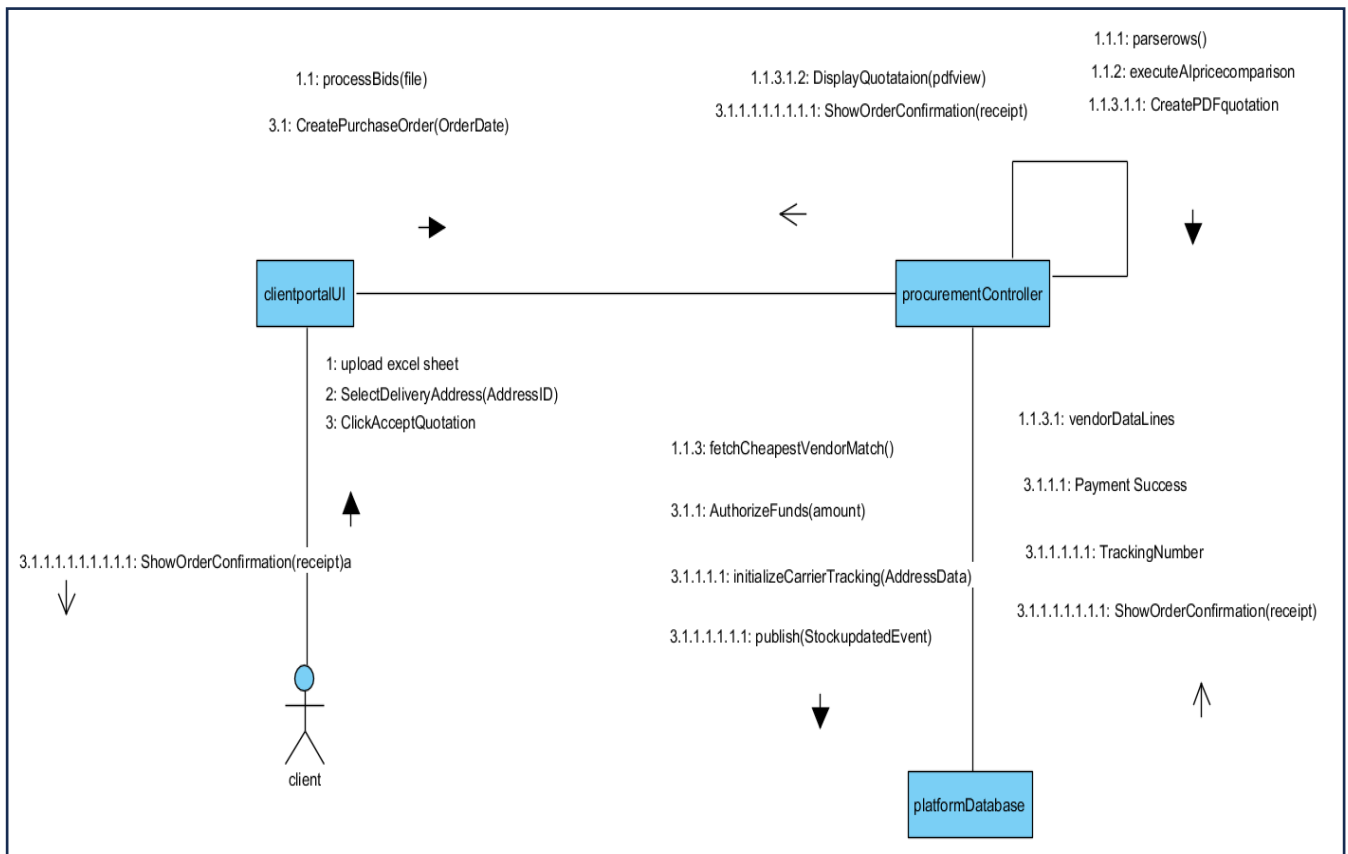
5.4 Sequence diagram – admin dashboard (staff)



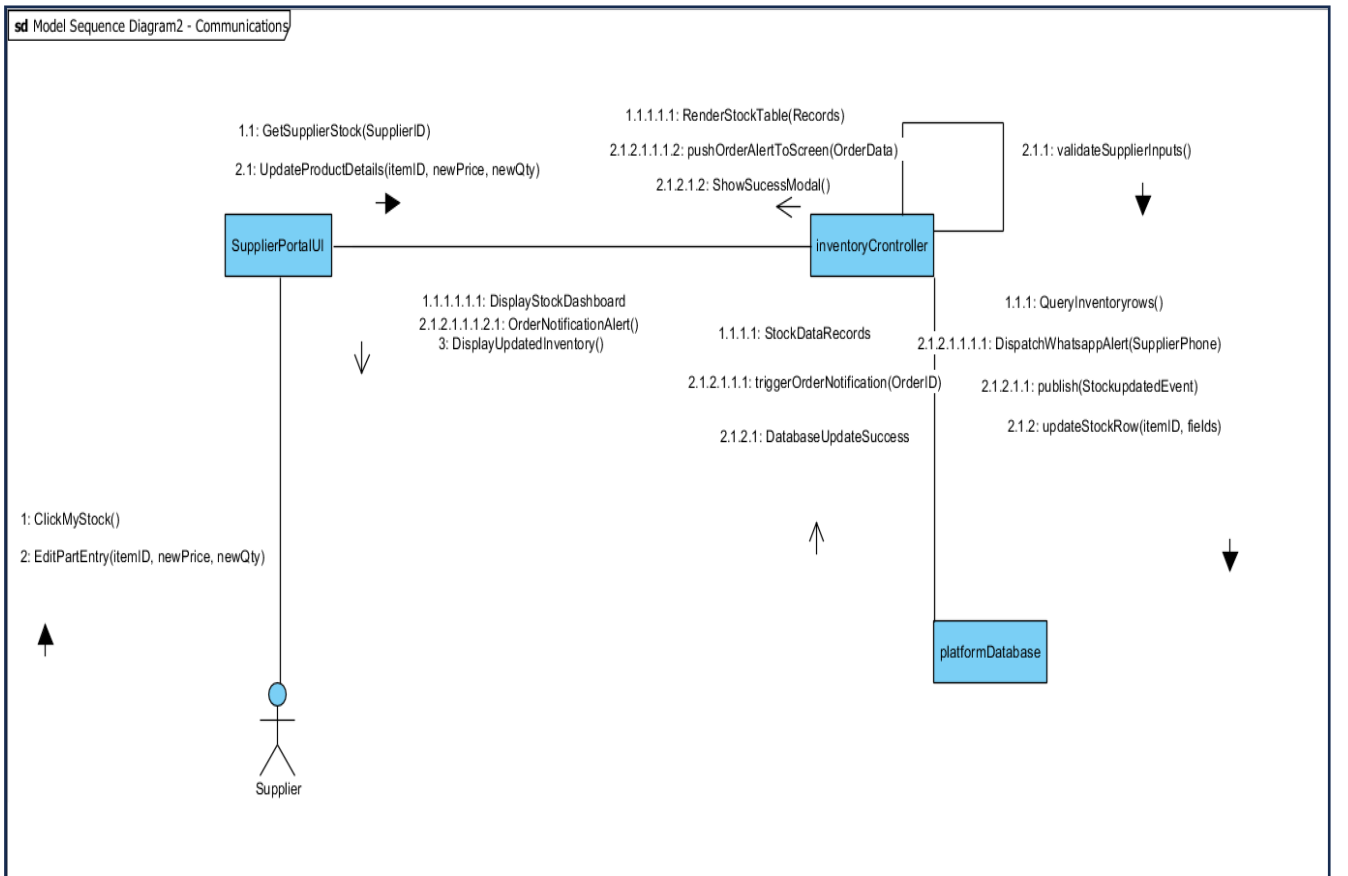
505 Sequence Diagram – integration gate



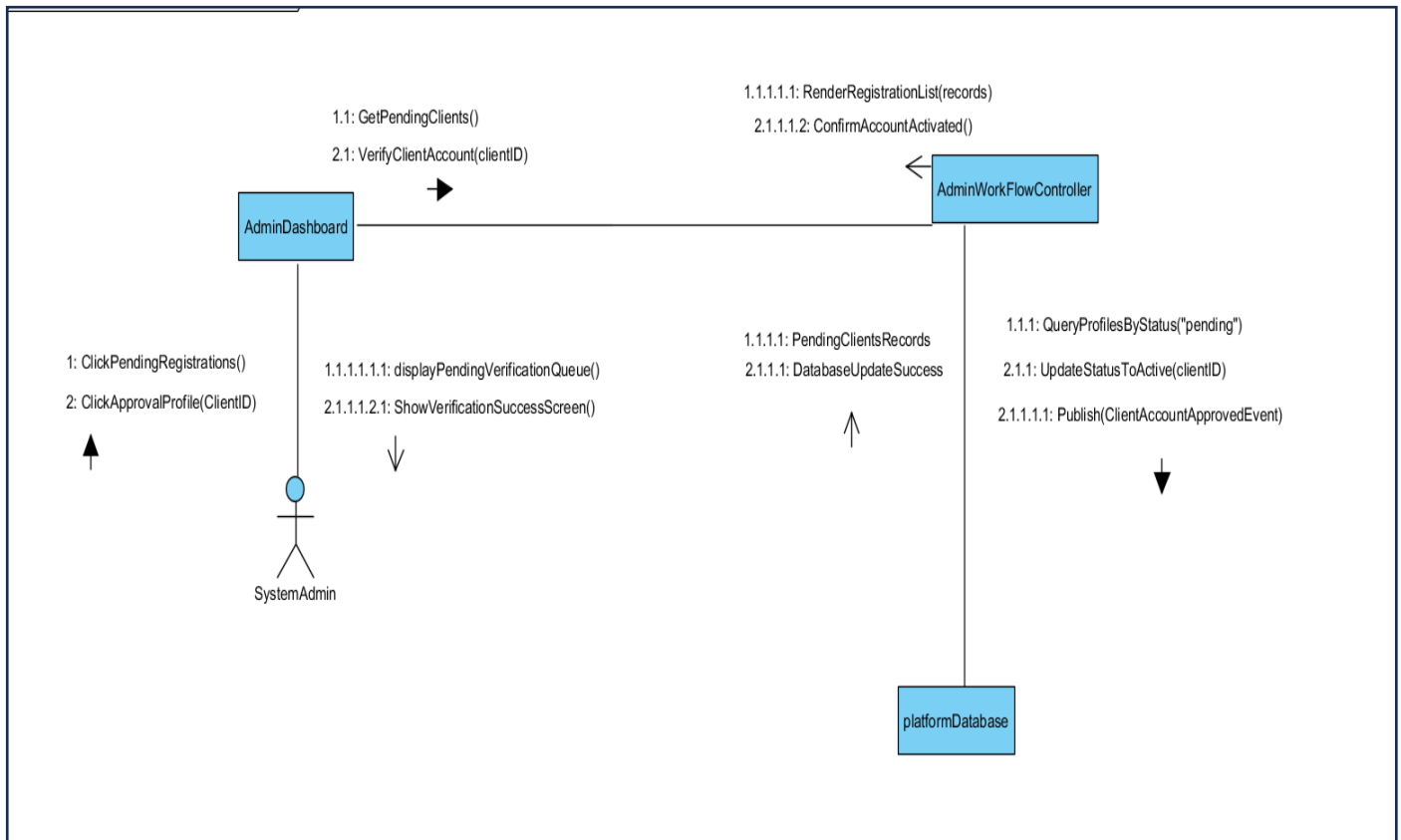
6.1 Communication diagram – client portal



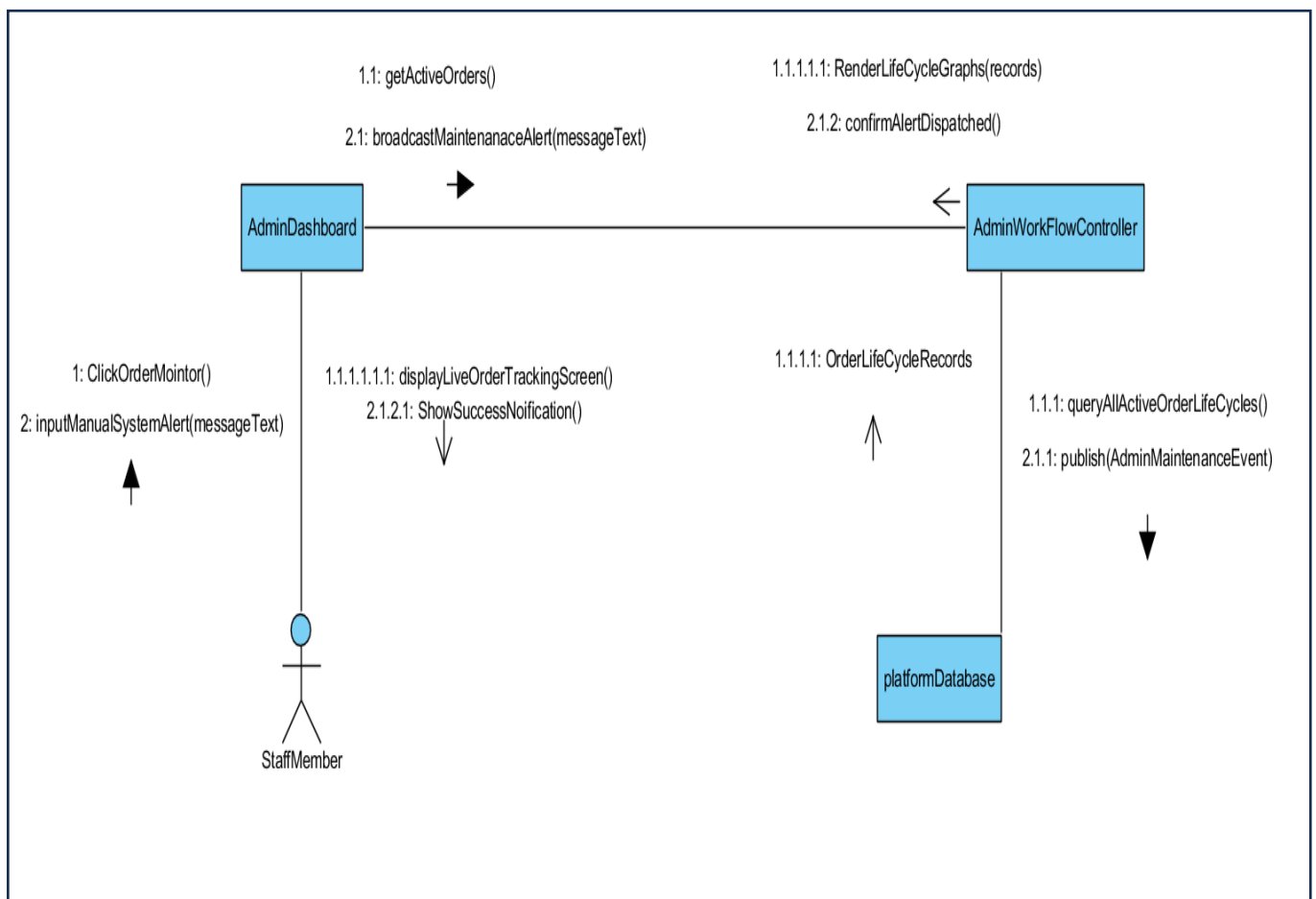
6.2 communication diagram – supplier portal



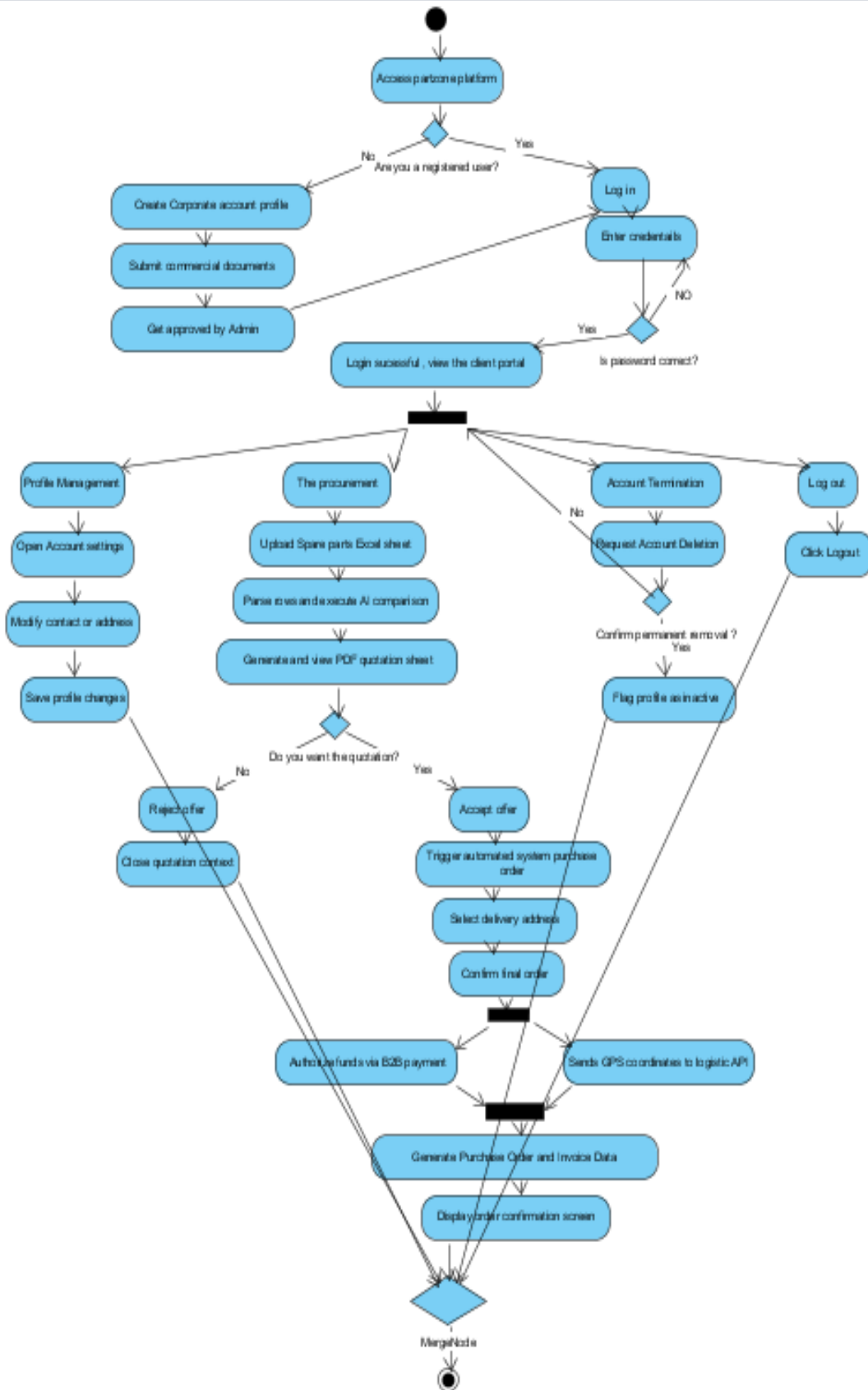
6.3 Communication diagram – Admin dashboard (admin)



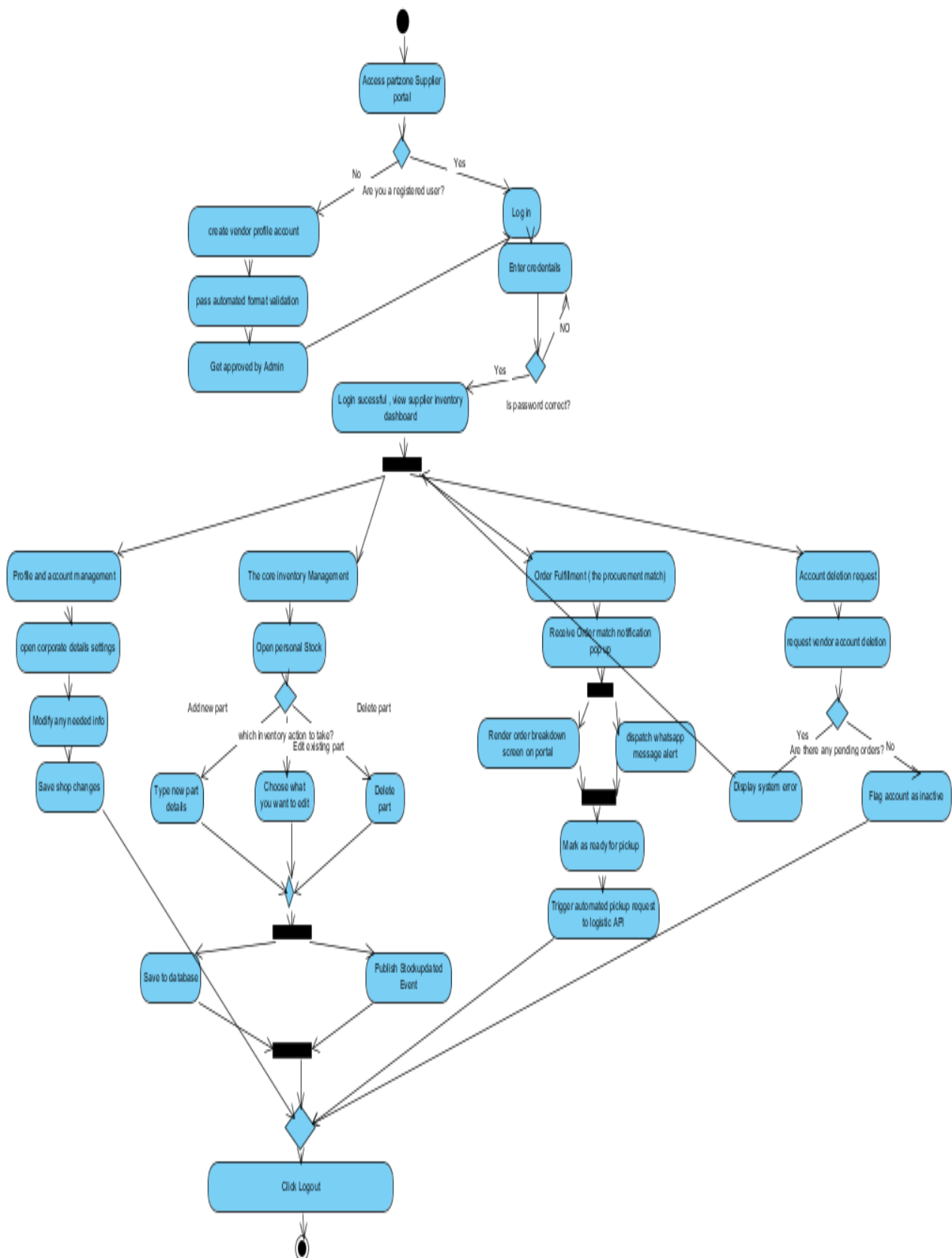
6.4 communication diagram – admin dashboard (staff)



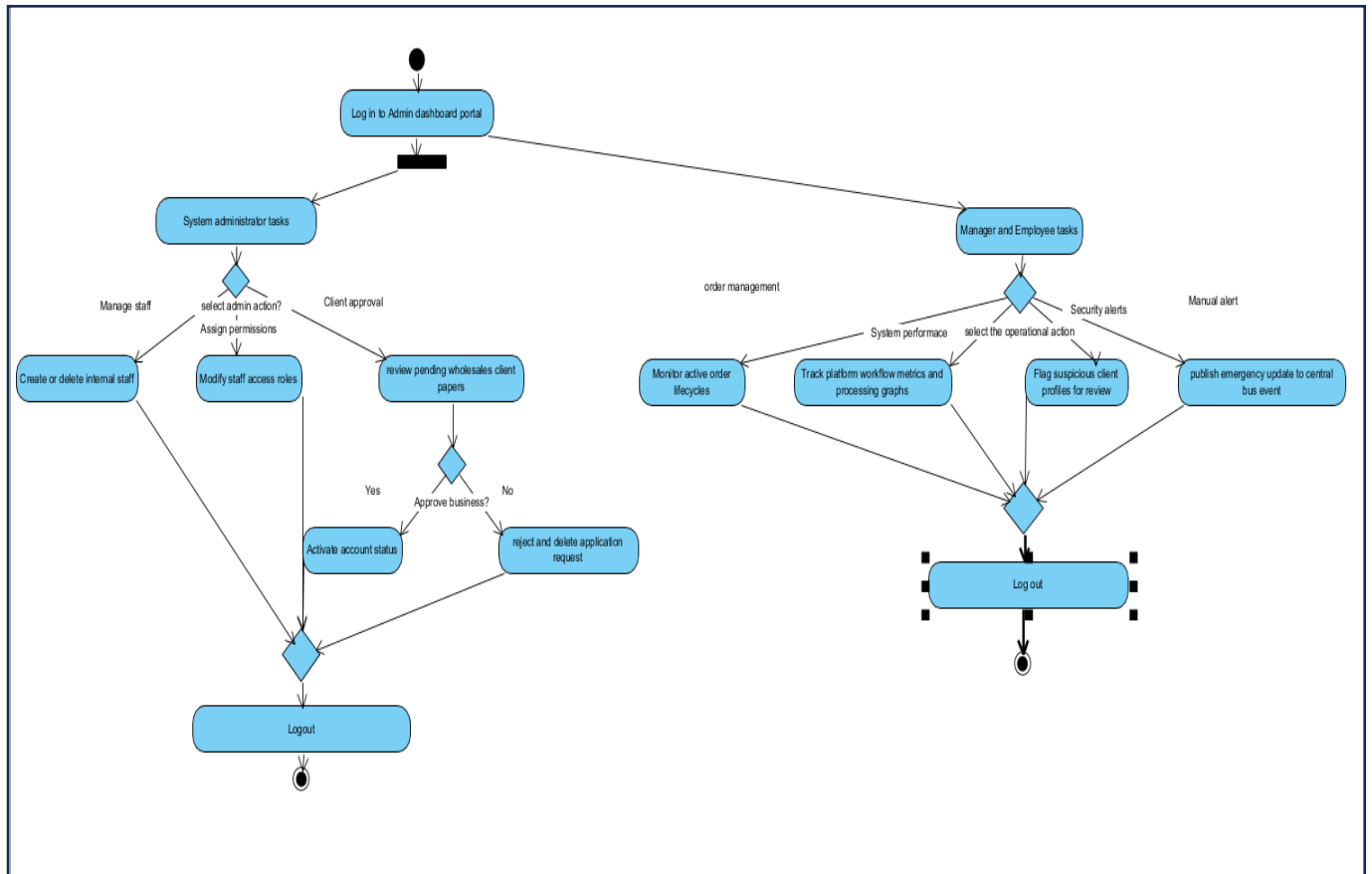
7.1 Activity diagram – client portal



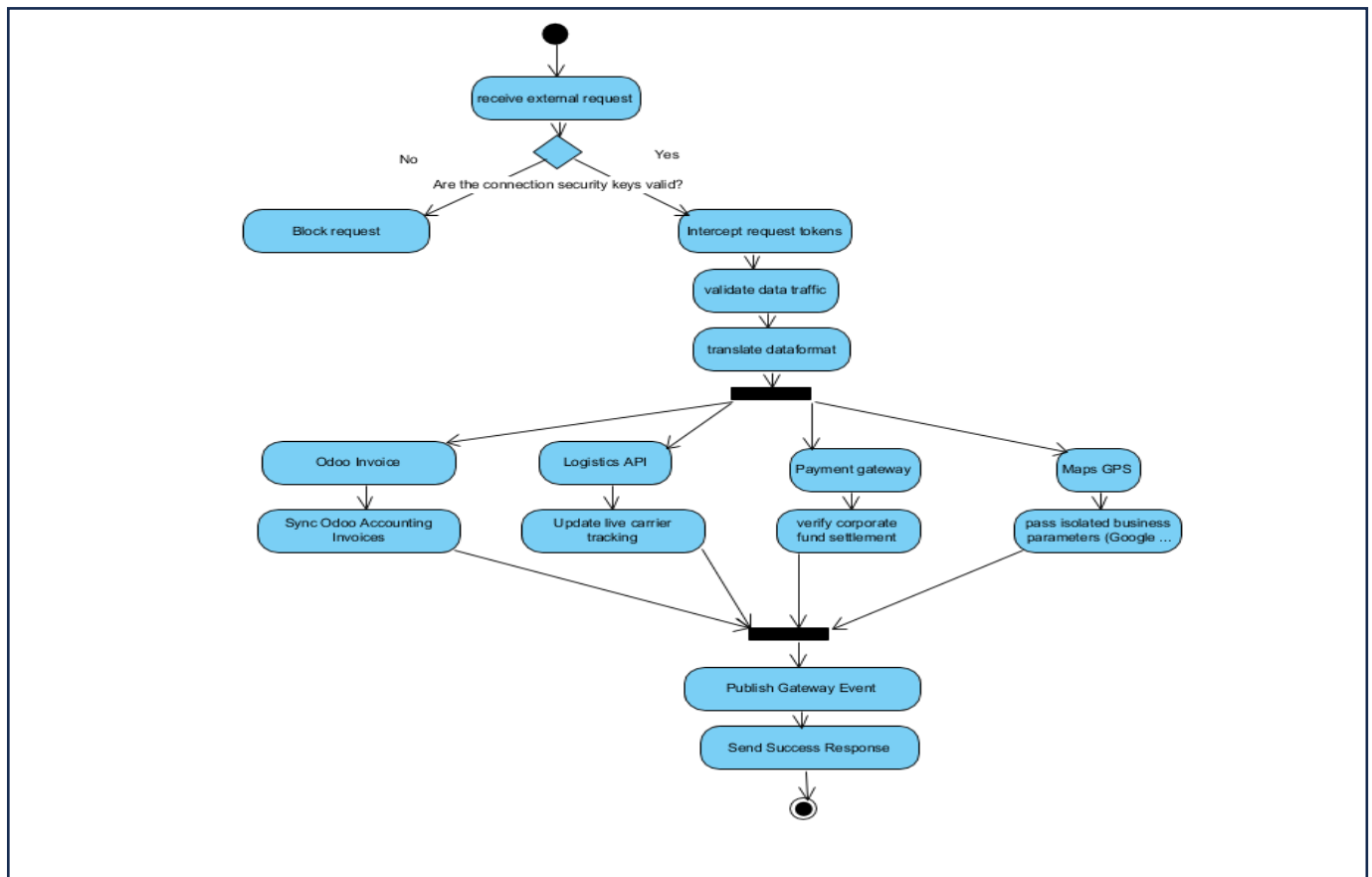
7.2 Activity diagram – supplier portal



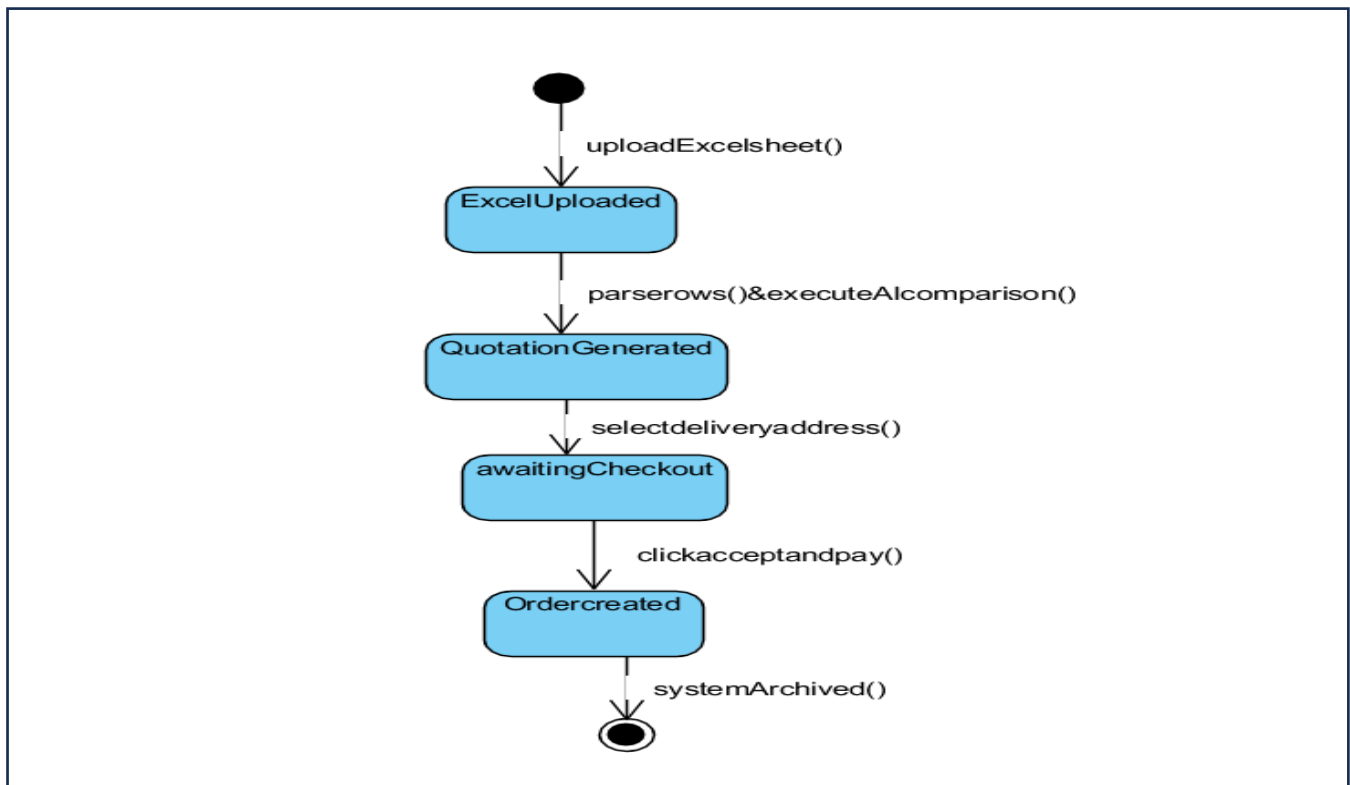
7.2 Activity diagram – admin dashboard



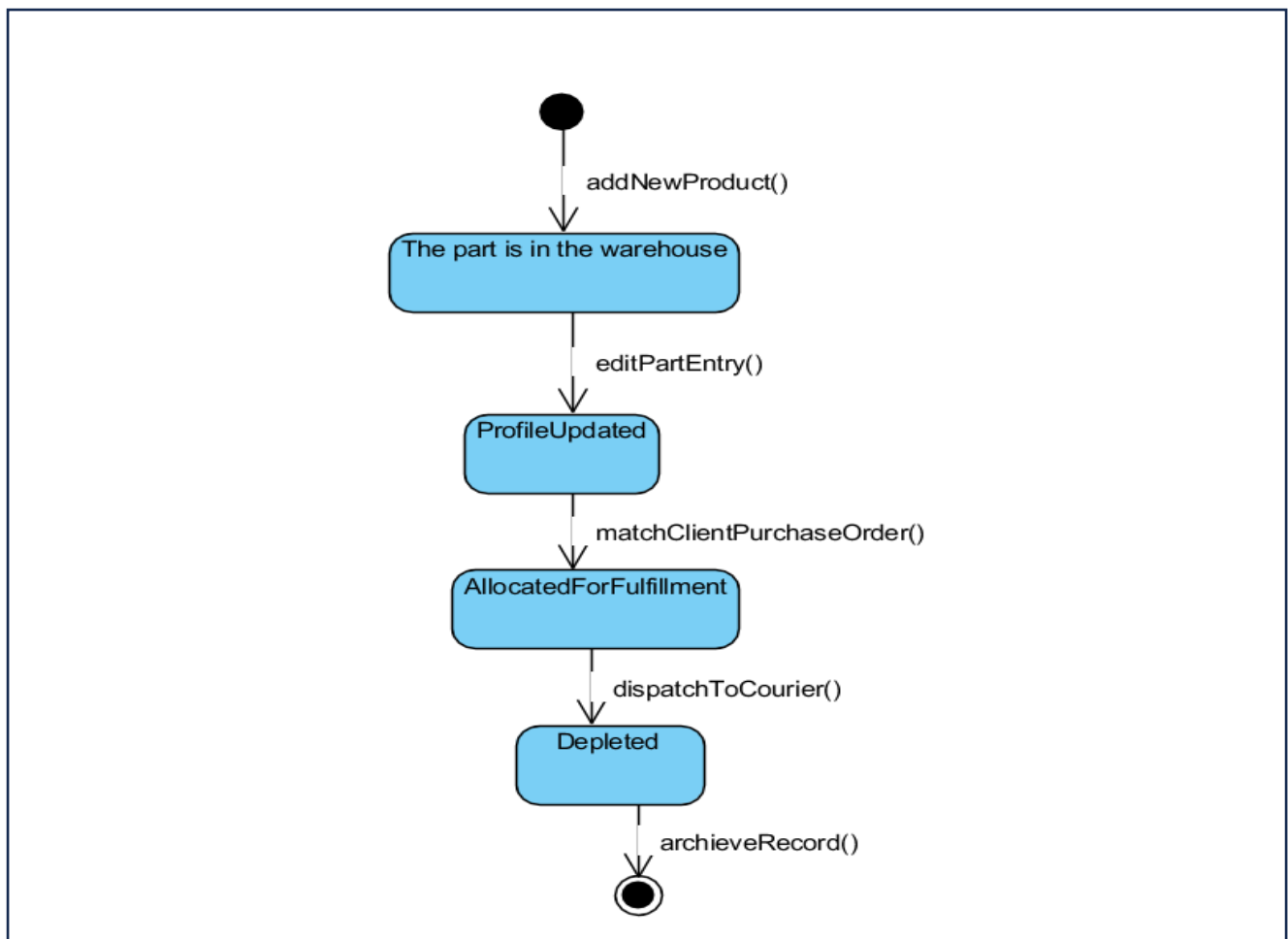
7.4 Activity diagram – integration gate



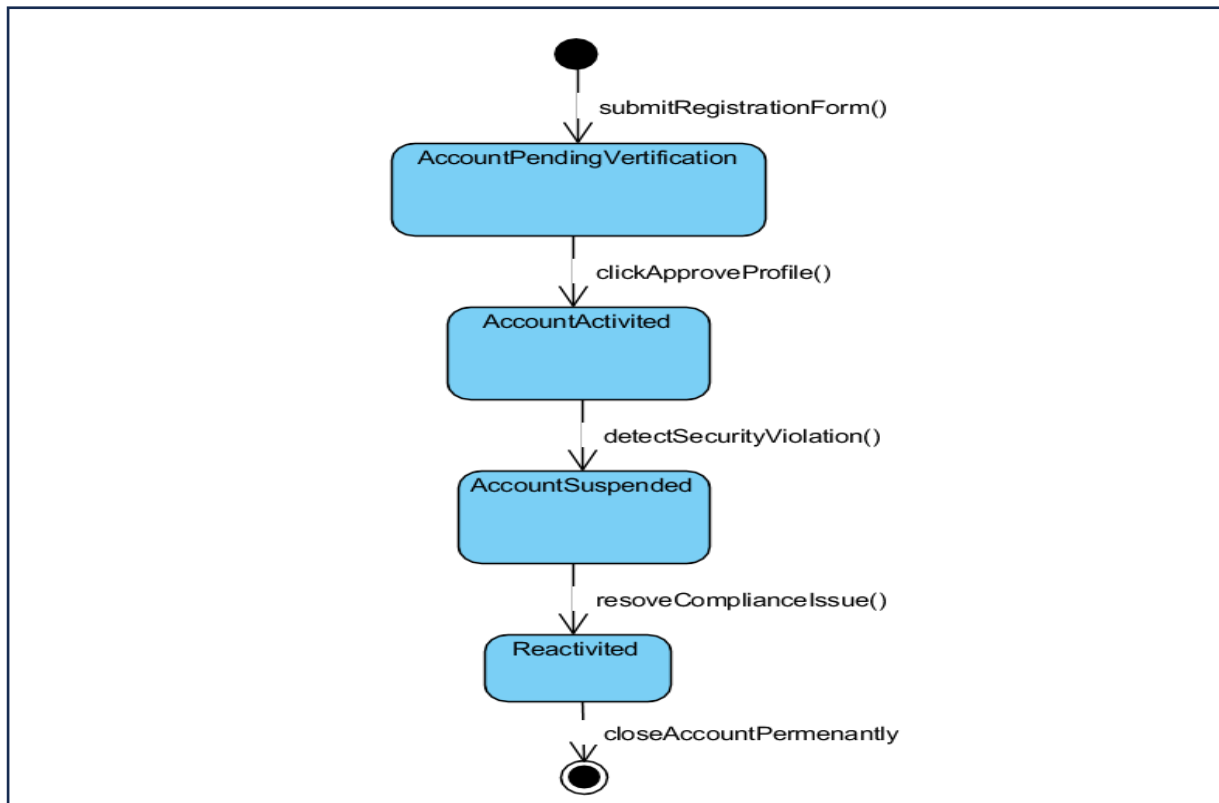
8.1 state machine diagram – client portal (wholesale quotation entity)



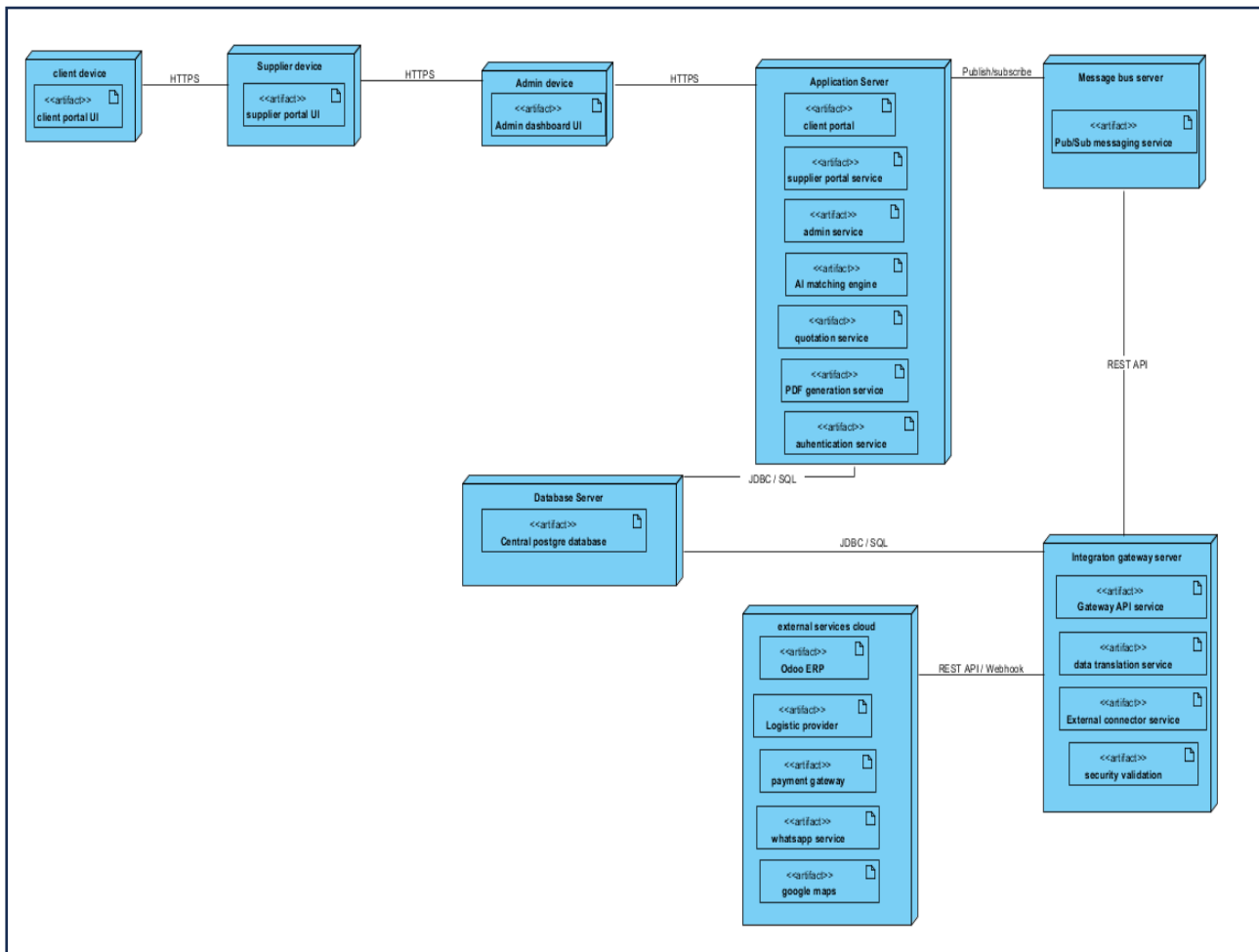
8.2 state machine diagram – supplier portal (Inventory stock entity)



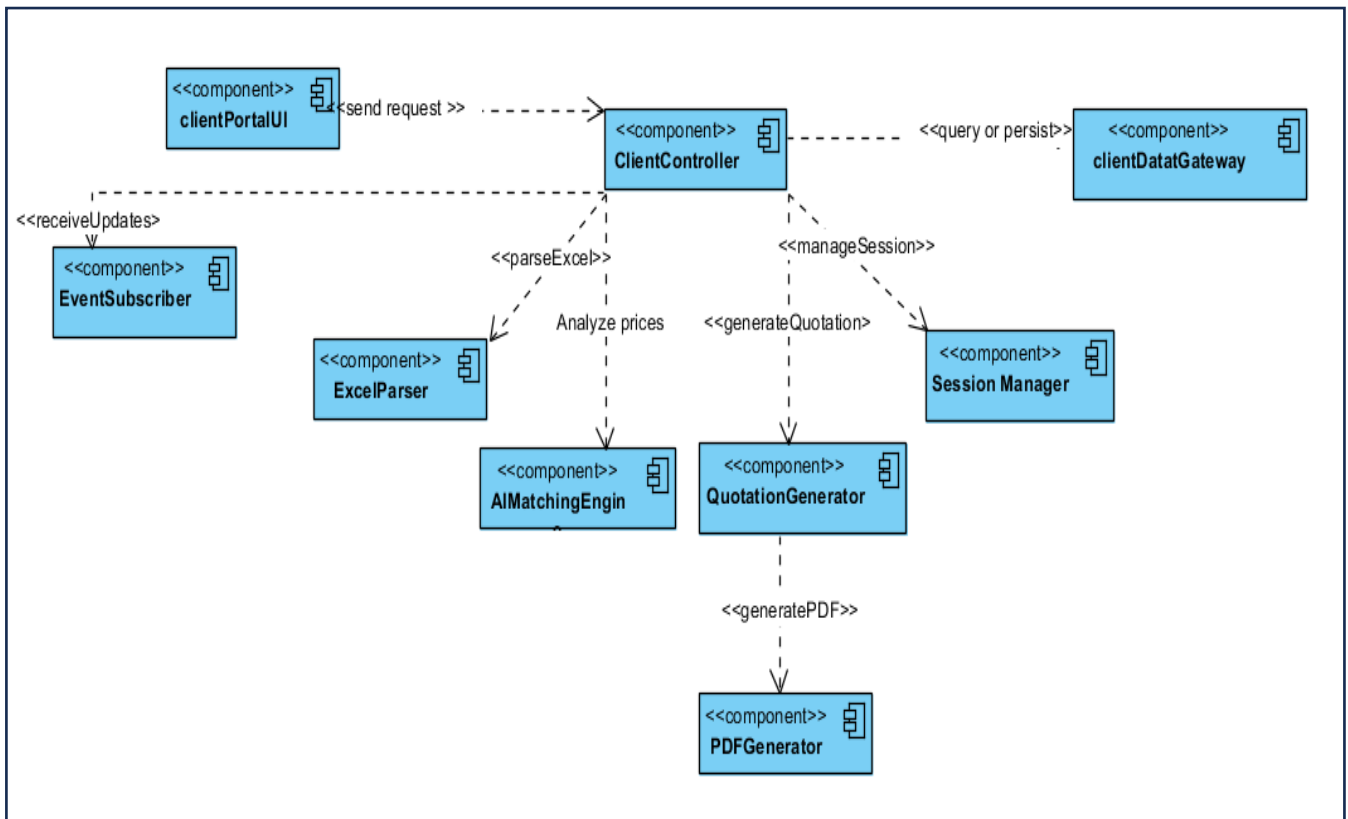
8.3 state machine diagram – Admin dashboard (Client user account entity)



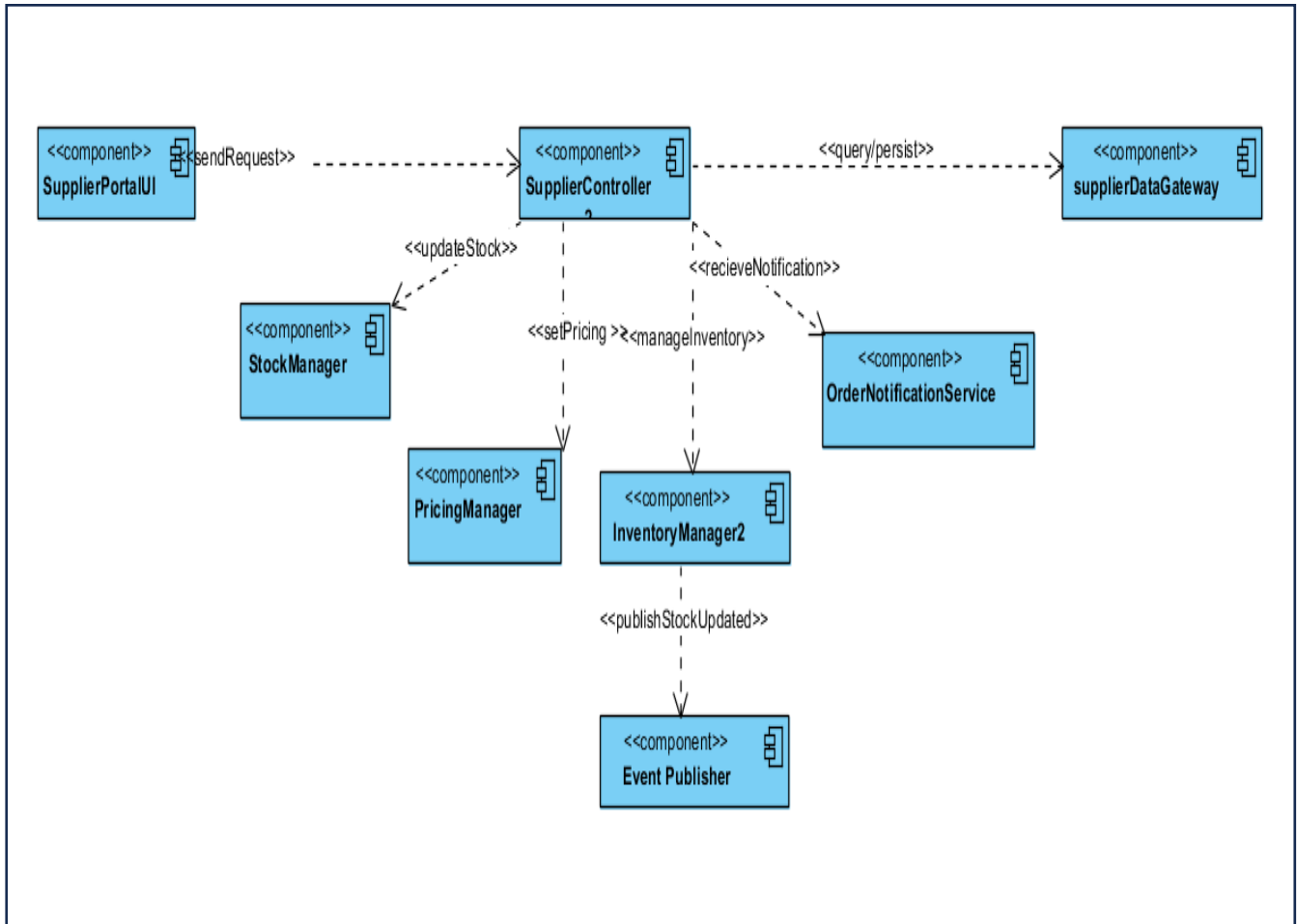
9 – Deployment diagram



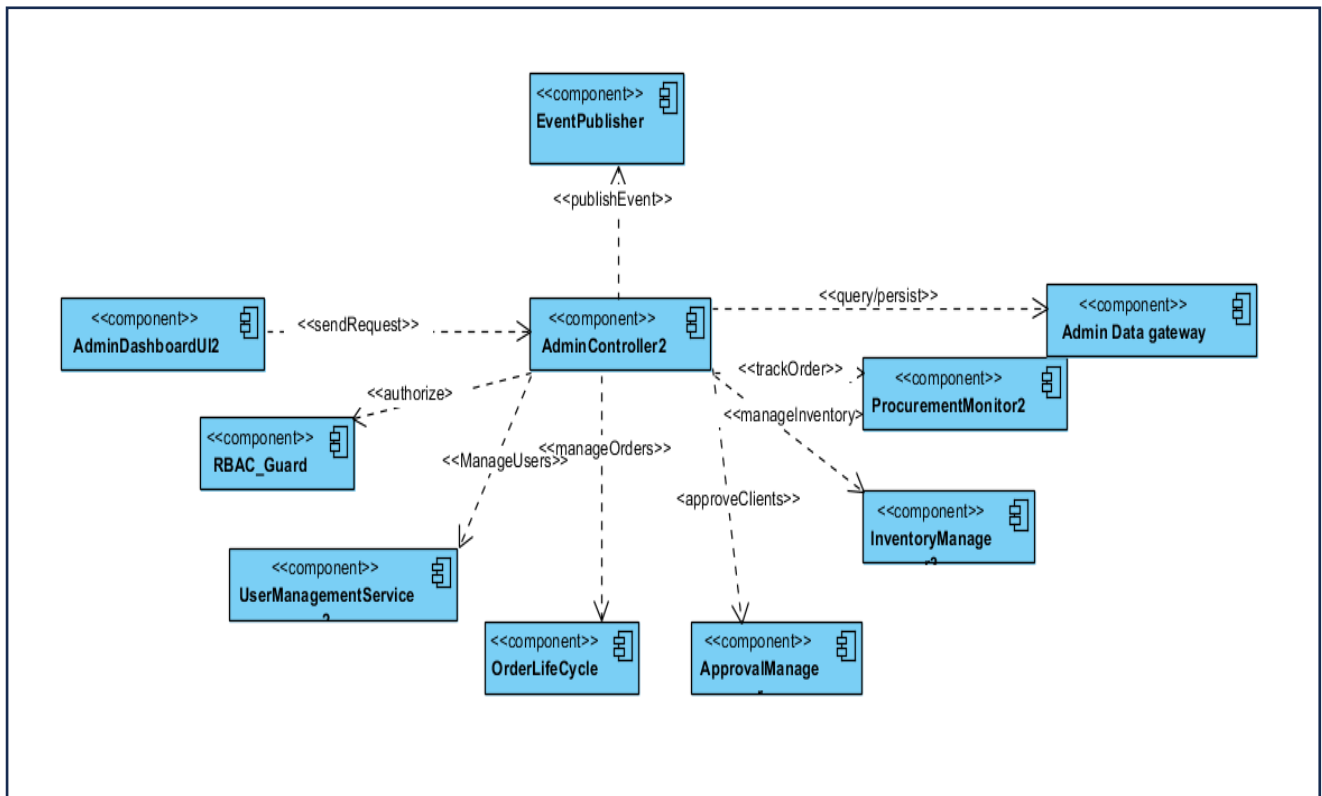
10.1 component diagram (client)



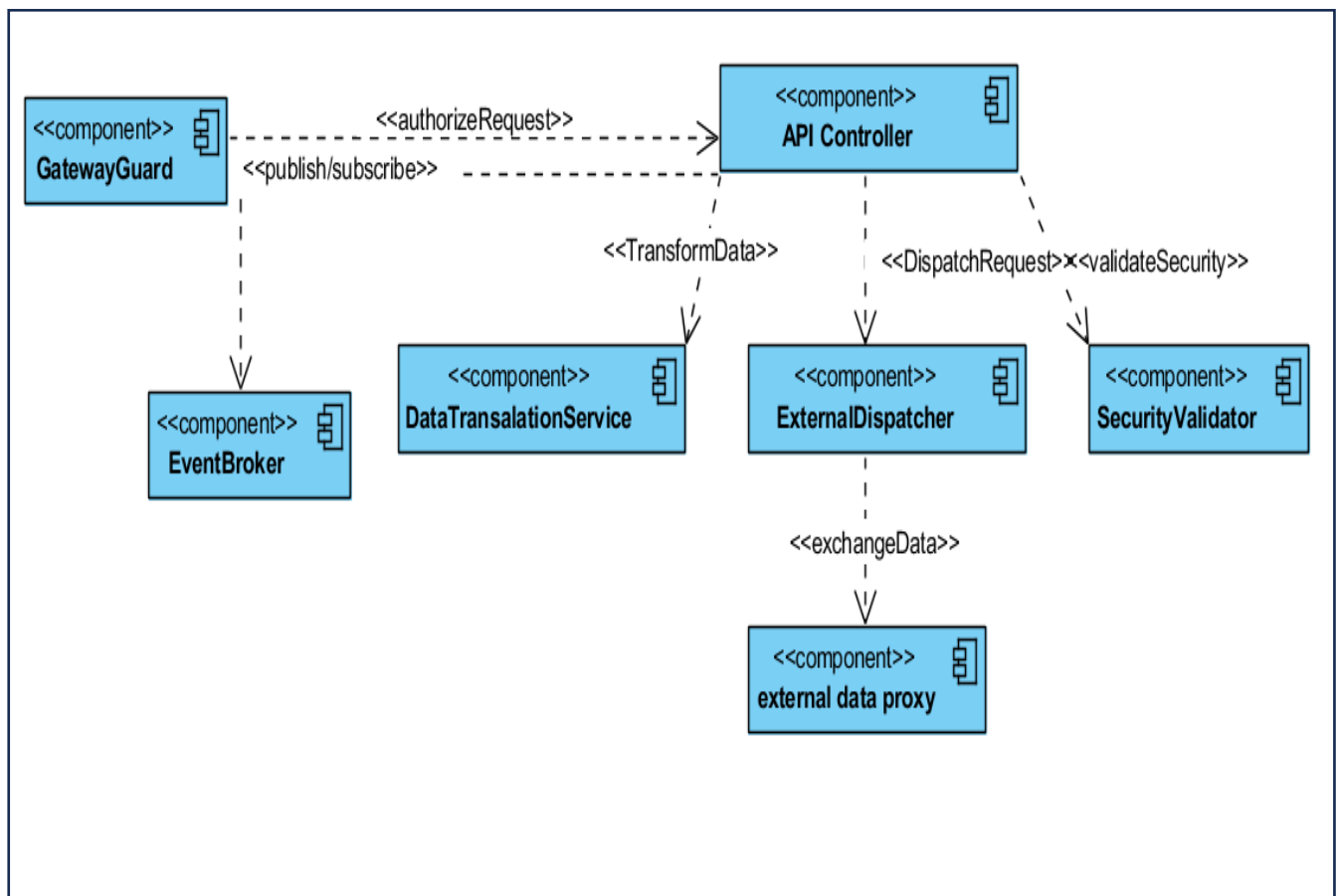
10.2 component diagram (supplier)



10.3 component diagram (admin)



10.4 component diagram – integration gate



Phase 3: Architecture and structure of the system

3.1 Architectural Analysis & Component Specification

1- Client Portal Components

The Client Portal is used by wholesale customers to create accounts, upload spare parts request files, compare prices, and receive quotations.

- **ClientPortalUI:** The main interface used by clients to register, log in, upload Excel files, and view quotations.
- **ClientController:** Handles requests coming from the client interface and connects the portal with the system services.
- **ExcelParser:** Reads the uploaded Excel file and extracts the spare parts data.
- **AIMatchingEngine:** Compares prices between available suppliers and selects the best available prices for each requested item.
- **QuotationGenerator:** Creates the quotation after price comparison.
- **PDFGenerator:** Converts the quotation into a PDF file that can be downloaded or sent to the client.
- **SessionManager:** Maintains user login sessions and portal access.

2- Local Supplier Portal Components

The Supplier Portal allows local suppliers to manage stock, prices, and receive order updates.

- **SupplierPortalUI:** Interface used by suppliers to manage products and prices.
- **SupplierController:** Handles supplier requests and inventory operations.
- **InventoryManager:** Manages stock quantity and product availability.
- **PricingManager:** Handles product pricing updates.
- **OrderNotificationService:** Sends notifications when an order is accepted.
- **EventPublisher:** Publishes stock update events to keep the system synchronized.

3- Admin Dashboard Components

The Admin Dashboard is used to manage users, approvals, inventory, and system operations.

- **AdminDashboardUI:** Interface used by admins and employees to manage system operations.

- **AdminController:** Handles admin requests and system management tasks.
- **RBACGuard:** Controls permissions and checks user roles before allowing access.
- **UserManagementService:** Manages client and supplier accounts.
- **InventoryManager:** Allows management of inventory information.
- **OrderLifecycleManager:** Monitors order status and updates.
- **ApprovalManager:** Handles approval of new wholesale client accounts.

4- Integration Gateway Components

The Integration Gateway connects PartZone with external systems and services.

- **GatewayGuard:** Verifies security before accepting external requests.
- **APIController:** Handles communication between PartZone and external systems.
- **DataTranslationService:** Converts external data into a format used by the system.
- **ExternalDispatcher:** Sends and receives data from external services.
- **EventBroker:** Supports communication between services using events.
- **SecurityValidator:** Ensures safe data exchange with external systems

3.2 Internal Layered Architecture

1- Client Portal Architecture

The Client Portal follows a layered structure to separate user interface, business logic, and data handling.

- **Presentation Layer:** Contains ClientPortalUI for user interaction and file uploads.
- **Business Logic Layer:** Contains ClientController, ExcelParser, AIMatchingEngine, QuotationGenerator, PDFGenerator, and SessionManager to process requests and generate quotations.
- **Data Access Layer:** Handles communication with the database and system services.

2- Supplier Portal Architecture

The Supplier Portal uses a layered structure to manage stock updates and supplier operations.

- **Presentation Layer:** Contains SupplierPortalUI.

- **Business Logic Layer:** Contains SupplierController, InventoryManager, PricingManager, OrderNotificationService, and EventPublisher.
- **Data Access Layer:** Handles supplier-related database communication.

3- Admin Dashboard Architecture

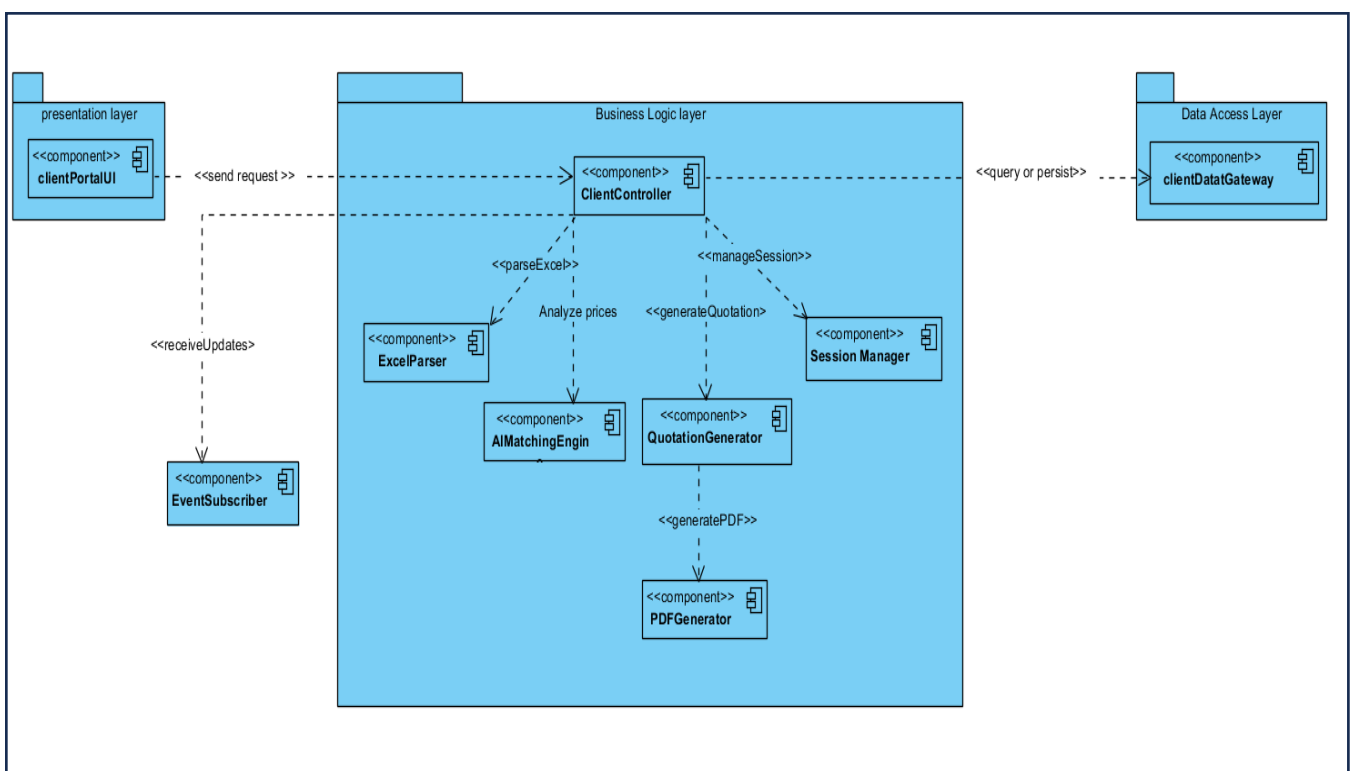
The Admin Dashboard is designed to securely manage system activities.

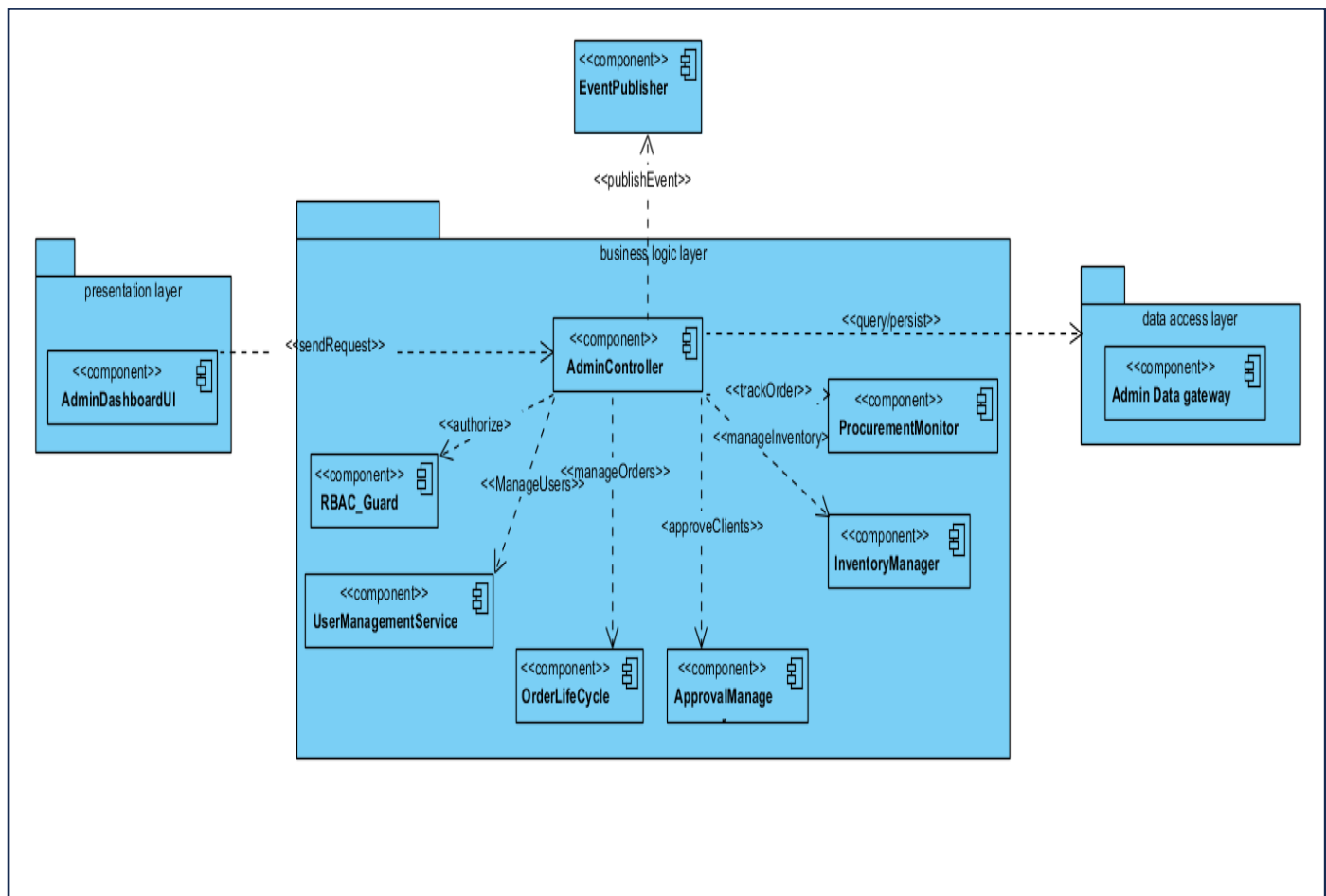
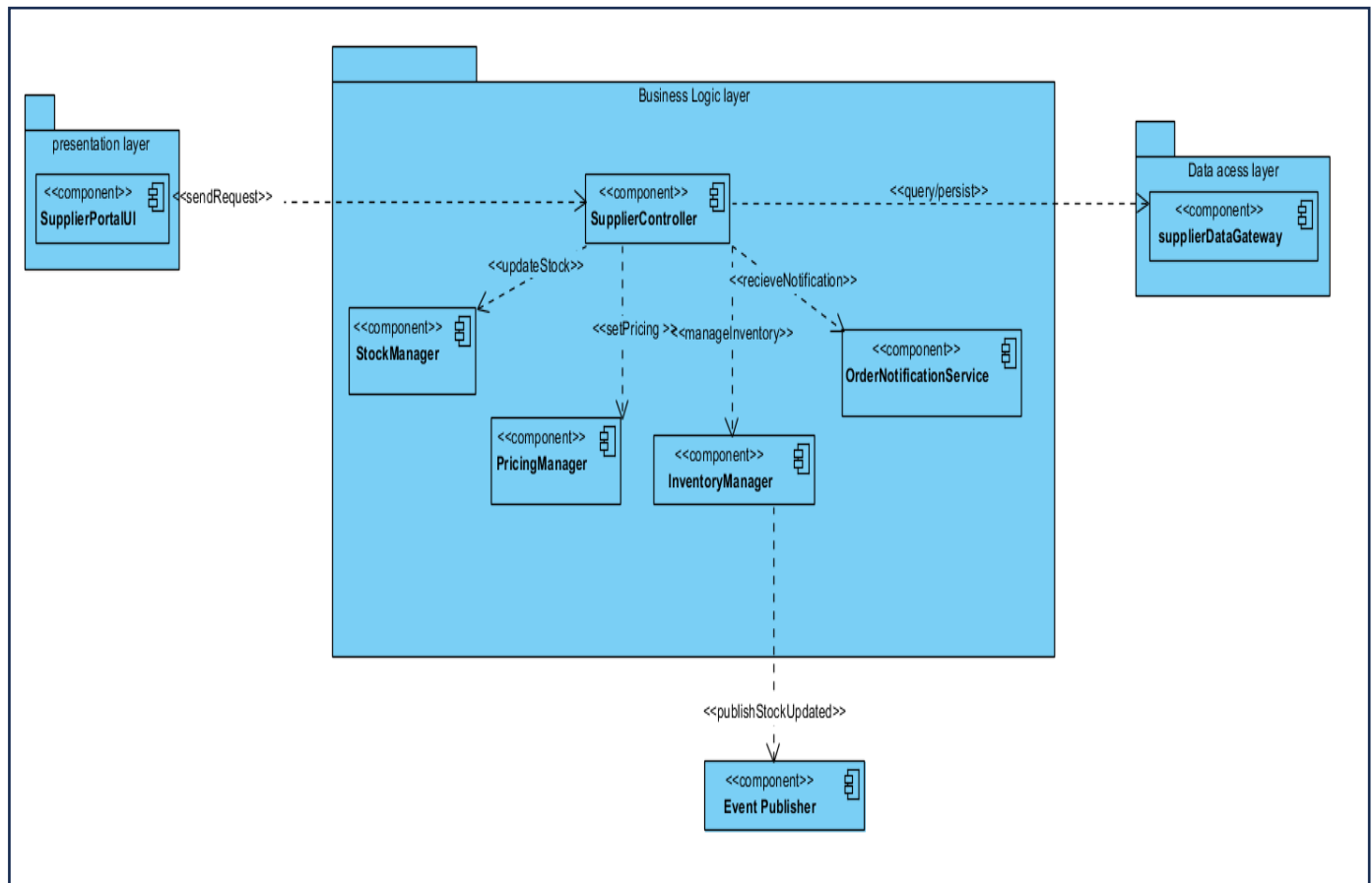
- **Presentation Layer:** Contains AdminDashboardUI.
- **Business Logic Layer:** Contains AdminController, RBACGuard, UserManagementService, InventoryManager, OrderLifecycleManager, and ApprovalManager.
- **Data Access Layer:** Handles communication with the database and stores system updates.

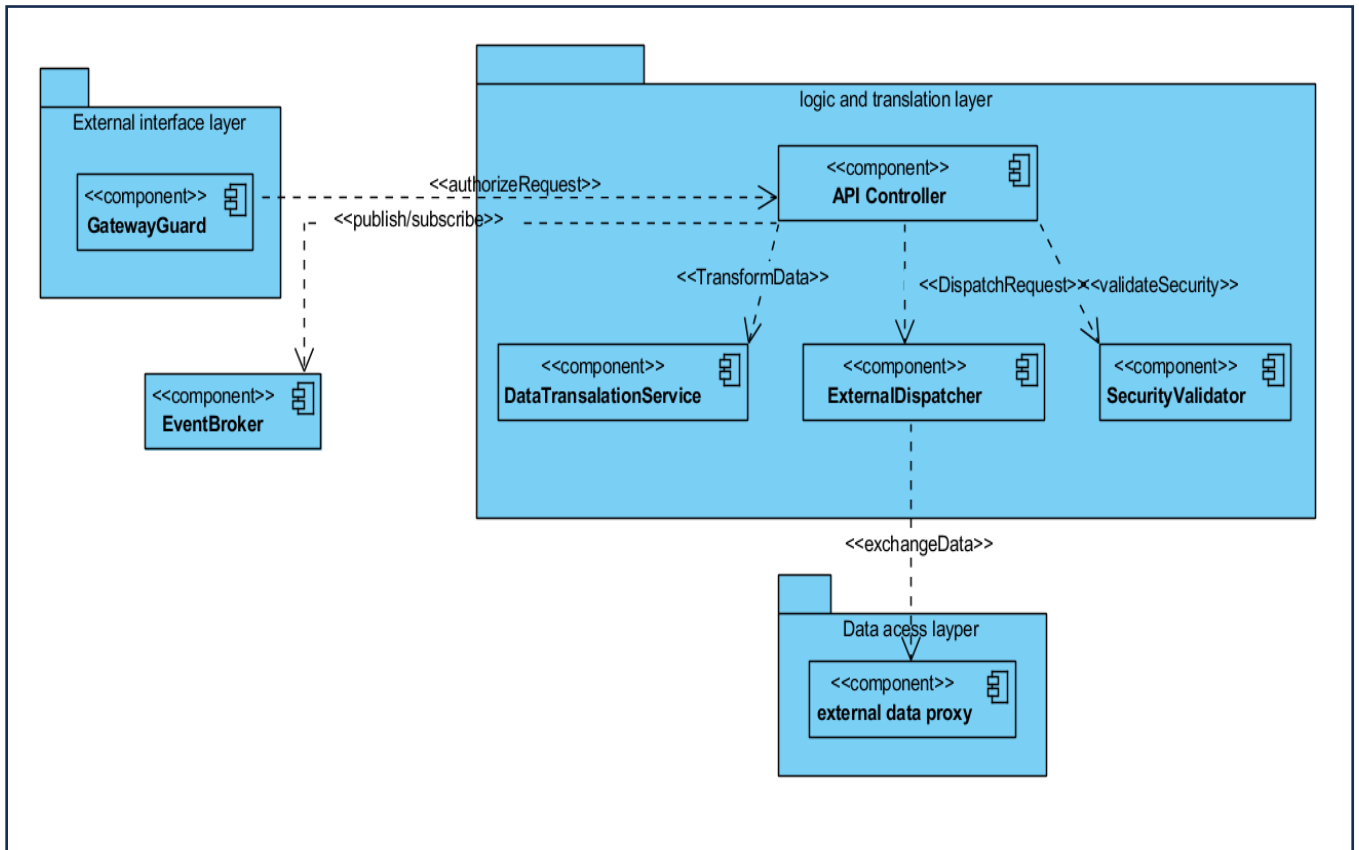
4- Integration Gateway Architecture

The Integration Gateway manages communication with external systems.

- **External Interface Layer:** Contains GatewayGuard and APIController for handling external requests.
- **Logic & Transformation Layer:** Contains DataTranslationService, ExternalDispatcher, EventBroker, and SecurityValidator.
- **Data Access Layer:** Handles data exchange between external systems and the main database.







3 System Integration & Architectural Strategy

To ensure that all parts of the PartZone B2B platform work together correctly, the system follows an integrated architecture based on N-Tier Architecture and Event-Driven Communication (Pub/Sub). This design helps the system stay organized, scalable, and easy to manage while allowing all portals to communicate efficiently.

A. Event-Based Communication (Pub/Sub)

PartZone uses an Event Bus to connect different parts of the system without requiring direct communication between portals.

This approach improves system performance and reduces dependency between system modules. It also supports real-time updates and helps the platform handle many users at the same time.

B. Centralized Database Integration

All portals in the system connect to one Central PostgreSQL Database. This database stores important information such as user accounts, quotations, orders, inventory, payments, and shipment records.

Although each portal performs different tasks, all final operations are saved in the same database to maintain data consistency and prevent conflicts between system modules.

For example:

- The Client Portal stores quotation requests and accepted orders.
- The Supplier Portal updates stock availability and pricing.
- The Admin Dashboard manages approvals, users, and system monitoring.
- The Integration Gateway exchanges information with external systems.

Using one centralized database ensures that all portals work with the same updated information.

C. Integration with External Systems

The Integration Gateway acts as a bridge between PartZone and external services.

It is responsible for receiving and sending data to third-party systems.

Before any external data enters the system, the gateway validates and processes it to make sure it matches the system requirements.

D. Layered System Architecture

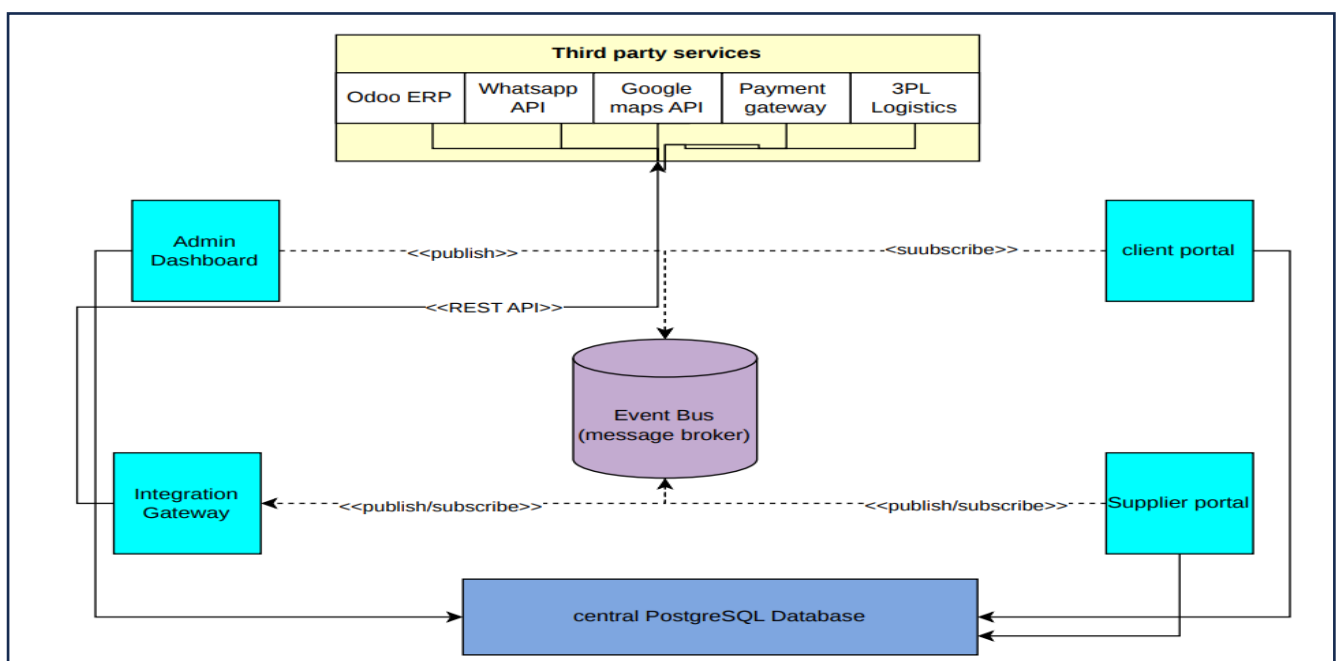
Each portal in PartZone follows a layered architecture to separate responsibilities inside the system.

The architecture consists of:

- Presentation Layer: Responsible for user interaction and interfaces.
- Business Logic Layer: Responsible for processing system operations and business rules.
- Data Access Layer: Responsible for database communication and data storage.

This separation makes the system easier to maintain and allows updates to one part of the system without affecting the others.

E. The system integration diagram that illustrates the previous explanation:



4 The master Architecture Diagram:

